



An adaptive interval many-objective evolutionary algorithm with information entropy dominance

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ABSTRACT

Interval many-objective optimization problems (IMaOPs) involve more than three conflicting objectives with interval parameters. Various real-world applications under uncertainty can be modeled as IMaOPs to solve, so effectively handling IMaOPs is crucial for solving practical problems. This paper proposes an adaptive interval many-objective evolutionary algorithm with information entropy dominance (IMEA-IED) to tackle IMaOPs. Firstly, an interval dominance method based on information entropy is proposed to adaptively compare intervals. This method constructs convergence entropy and uncertainty entropy related to interval features and innovatively introduces the idea of using global information to regulate the direction of local interval comparison. Corresponding interval confidence levels are designed for different directions. Additionally, a novel niche strategy is designed through interval population partitioning. This strategy introduces a crowding distance increment for improved subpopulation comparison and employs an updated reference vector method to adjust the search regions for empty subpopulations. The IMEA-IED is compared with seven interval optimization algorithms on 60 interval test problems and a practical application. Empirical results affirm the superior performance of our proposed algorithm in tackling IMaOPs.

1. Introduction

Over the past two decades, the field of evolutionary algorithms (EAs) [1–3] has seen significant advancements, with a plethora of methodologies proposed to address multi-objective optimization problems (MOPs) [4–6] and their more complex counterparts, many-objective optimization problems (MaOPs) [7–9]. These algorithms have been instrumental in finding solutions [10] to various optimization challenges [11] across various disciplines. However, a critical limitation emerges when considering the inherent uncertainties present in real-world optimization problems [12]. Such uncertainties, often manifesting as uncertain parameters [13–15], can significantly affect the reliability and robustness of the optimization outcomes. These parameters can be categorized into three primary types: fuzzy parameters [16–18], random parameters [19–21], and interval parameters [22–24]. Among these, interval parameters offer a pragmatic approach to measuring uncertainty, given that determining the precise membership functions for fuzzy numbers or probability distributions

for random variables can be both theoretically challenging and practically demanding [25]. Conversely, establishing the upper and lower bounds of intervals, along with the interval radius and midpoint, is a more feasible task. The presence of interval parameters transforms MaOPs into a specialized category known as interval many-objective optimization problems (IMaOPs). IMaOPs has applications in diverse fields, including interval many-objective cloud task scheduling [26], unmanned aerial vehicles (UAVs) oilfield inspection problem [27], and workflow migration in edge computing environments [28]. Despite the effectiveness of traditional EAs such as NSGA-III [29], MOEA/DD [30], and KnEA [31] in handling deterministic MaOPs, they fall short when faced with the complexities of IMaOPs. This highlights the need for developing novel approaches specifically tailored to solve IMaOPs.

At present, some interval evolutionary algorithms have been proposed to address IMOPs and IMaOPs. These methods can be categorized into two types. The first type converts the interval optimization problems to deterministic optimization problems. Subsequently,

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conventional deterministic evolutionary algorithms are applied to address the transformed problems. Bhunia et al. transform IMOP into a single-objective optimization problem using interval order relations and interval metrics [32]. Guo et al. proposed a knowledge-guided MOEA/D using interval midpoints [33]. Gong et al. integrated the midpoint and width of intervals to transform the interval objective into a precision objective, then applied an evolutionary algorithm to solve the transformed optimization problem [34]. SetGA proposed by Gong et al. [35] uses an indirect method to convert IMAOPs into a bi-objective MOP with objectives of hypervolume and uncertainty and then uses a set-based genetic algorithm to solve the MOP. These methods fall under the category of indirect methods [12]. Obviously, such methods lose interval information, which has a negative impact on the accuracy of the original problem.

Our study focuses on the second type approach, which constructs interval dominance methods and performance metrics to address interval optimization problems directly. This methodology yields an interval Pareto front (PF) without compromising the inherent interval characteristics of the problem. In tackling interval optimization problems, balancing traditional optimization objectives with effectively managing interval uncertainty is crucial. The interval dominance method serves as a mechanism within the non-dominated sorting framework [36] to compare interval-valued solutions. This method enables the selection of more favorable solutions by comprehensively accounting for both uncertainty and convergence, thus enhancing decision-making in uncertain environments. For instance, Limburg et al. proposed an interval dominance method that comprehensively compares the convergence ability of upper and lower bounds [37]. But this method does not have uncertainty comparison, and when one interval covers the other, this method cannot compare two intervals. To increase the selection ability of interval dominance methods, Zhang et al. proposed a multi-objective particle swarm optimization algorithm based on probabilistic dominance [38]. Gong et al. proposed maximum interval confidence based on the distance to the ideal interval [25]. Liu et al. developed interval probability to compare two intervals [39]. Zhang et al. incorporate interval credibility into the artificial bee colony algorithm [40]. Lan et al. proposed additive and multiplicative dominance degrees by utilizing reference intervals [41]. These study implicitly considers interval convergence and uncertainty by calculating the dominance probability. However, the above dominance methods lack a more in-depth analysis of the special location relationships in intervals. Yi et al. considered the difference in the relative size of the average convergence level and convergence potential and further designed an interval dominance relationship [42]. Chen et al. defined a modified interval Pareto dominance by calculating three interval dominance probabilities for three types of location relationships [43]. However, the above different methods often result in different optimal interval solutions. To reduce the difficulty of selecting interval dominance methods, some studies have proposed ensemble methods. Sun et al. developed an assembled framework for interval dominance rules to compare different solutions [44]. Gan et al. proposed IMOEAD [12] and constructed an ensemble comparison of interval individuals utilizing the rank sum ratio method. However, as the number of interval objectives increases, the computational complexity of the ensemble method also increases. It remains necessary to improve interval dominance methods for selecting superior interval solutions. To compare two intervals with the same midpoint, Cui et al. proposed a conditional interval confidence dominance relation [45] that emphasizes lower bound convergence. However, intervals with good lower bound convergence often face poor interval uncertainty. Therefore, a detailed discussion is needed on the trade-off between convergence and uncertainty.

Above improved dominance methods enhance the ability to select intervals, but most of them only analyze the location relationship between two intervals and set fixed selection criteria. However, the population constantly changes in the evolution process [46], so static approaches may difficult to balance convergence and uncertainty. This

could potentially result in overlooking some better interval solutions for IMAOPs. It is necessary to propose an interval dominance method that can adaptively adjust selection criteria according to changes in the population information. In addition, information entropy [47], a quantitative indicator for measuring disorder, has a wide range of applications in statistics and information theory. The degree of disorder can indirectly evaluate the importance of a certain attribute to decision-making. By analyzing the characteristics of interval optimization problems and information entropy, corresponding interval information entropy can be constructed to quantify the relative importance of interval convergence and uncertainty, which is particularly important in interval comparison. Therefore, an information entropy dominance method with adaptive interval selection is proposed to balance convergence and uncertainty.

In addition, many interval objectives involved in IMAOPs make the solution space complex, and obtaining a set of well-distributed solutions is difficult. For MaOPs, NSGA-III and its improvement methods [48–50] can effectively enhance population diversity and distribution. However, they still suffer from insufficient local and global search capabilities [27] and are unable to associate reference vectors with interval individuals. Therefore, it is necessary to improve the NSGA-III to address IMAOPs. At present, there is relatively little research on the above content. Ge et al. [27] introduced the matter-element extension model to calculate the correlation between interval solutions and reference vectors. They then combined a local fruit fly optimization algorithm with NSGA-III to enhance searchability. But it did not solve the problem of insufficient niche selection ability. Therefore, this study proposes an adaptive niche selection strategy.

The main contributions of this paper are summarized as follows:

(1) This paper proposes an information entropy-based dominance method to compare intervals. This approach offers a flexible balance between convergence and uncertainty. Firstly, we compared the convergence and uncertainty entropy of the overall intervals to determine the focus direction. Then the interval confidence level of the corresponding direction was constructed to compare different individuals' intervals.

(2) This paper proposes an adaptive niche selection strategy. This strategy is suitable for solving interval optimization problems, where a crowding distance increment method is designed to compare different subpopulations and an updated reference vector method is used to enhance global search capability.

(3) This paper compares different interval evolution algorithms on many IMAOPs. The experiment also analyzed the impact of different confidence level thresholds on algorithm performance, which has been rarely discussed in previous research. In addition, comparisons were made specifically for different interval dominance methods to demonstrate the effectiveness of information entropy dominance methods. The comprehensive experimental results demonstrate the good performance of our proposed algorithm.

The remaining structure of this paper is as follows: Section 2 reviews preliminaries, Section 3 introduces the proposed algorithm framework and main methods, Section 4 analyzes the comparison results of this algorithm with other existing algorithms on multiple IMAOPs, and Section 5 summarizes the content of this paper and discusses future work.

2. Preliminaries

2.1. Interval Many-objective Optimization Problems (IMAOPs)

The mathematical model for IMAOPs can be expressed as follows:

$$\begin{aligned} \min f(\mathbf{x}, C) &= (f_1(\mathbf{x}, C_1), f_2(\mathbf{x}, C_2), \dots, f_m(\mathbf{x}, C_m))^T \\ \text{s.t. } \mathbf{x} &\in \mathbb{R}^n, C_i = (C_{i1}, C_{i2}, \dots, C_{ih})^T, C_{ih} = [C_{ih}, \overline{C_{ih}}] \\ i &= 1, 2, \dots, m, h = 1, 2, \dots, l. \end{aligned} \quad (1)$$

This model represents a minimization problem, where $\mathbf{x} = (x_1, x_2, \dots, x_n)$ is a n -dimensional decision vector, $f(\mathbf{x}, C)$ is a m -dimensional objective function vector ($m > 3$). $f_i(\mathbf{x}, C_i)$ is a component of $f(\mathbf{x}, C)$, representing i th objective function with interval parameters. C_i is an h -dimensional interval parameter vector, and its component C_{ih} is an interval. C_{ih} represents the lower bound of the interval, and $\bar{C}_{ih}$ represents the upper bound of the interval.

2.2. Interval arithmetic operations

The arithmetic operations for two intervals, $a = [\underline{a}, \bar{a}]$, $b = [\underline{b}, \bar{b}]$, are as follows [12]:

$$\begin{aligned} a + b &= [\underline{a} + \underline{b}, \bar{a} + \bar{b}] \\ a - b &= [\underline{a} - \bar{b}, \bar{a} - \underline{b}] \\ a \cdot b &= [\min S, \max S] \\ S &= \{ab, a\bar{b}, \bar{a}b, \bar{a}\bar{b}\} \\ a/b &= a \cdot (1/b) \end{aligned} \quad (2)$$

$$\begin{aligned} 1/b &= [\min(1/\bar{b}, 1/\underline{b}), \max(1/\bar{b}, 1/\underline{b})], 0 \notin b \\ \lambda a &= [\min(\lambda \underline{a}, \lambda \bar{a}), \max(\lambda \underline{a}, \lambda \bar{a})] \end{aligned}$$

For an interval vector $a = (a_1, a_2, \dots, a_n)$, its norm is defined as follows:

$$\begin{aligned} \|a\| &= \max_i |a_i|, i = 1, 2, \dots, n \\ |a_i| &= \max_i (|\underline{a}_i|, |\bar{a}_i|) \end{aligned} \quad (3)$$

$|\underline{a}_i|$ is equal to the absolute value of $\underline{a}_i$.

2.3. The Interval Dominance Methods (IDMs)

2.3.1. IDM1 based on maximal interval confidence

$$P(a \leq b) = \frac{d(b, \beta)}{d(a, \beta) + d(b, \beta)} \quad (4)$$

$$d(b, \beta) = \max \left\{ |\underline{b} - \underline{\beta}|, |\bar{b} - \bar{\beta}| \right\} \quad (5)$$

$$\beta = [\underline{\beta}, \bar{\beta}], \underline{\beta} = \min [\underline{a}, \underline{b}, \underline{b}], \bar{\beta} = \min \left\{ [\underline{a}, \bar{a}, \bar{b}] / \underline{\beta} \right\} \quad (6)$$

This method constructs an ideal interval β between two intervals and calculates the distances from each interval to the ideal interval. The closer an interval is to the ideal interval, the better it is. If $P > 0.5$, it indicates that a is closer to β , thus a dominates b . If $P < 0.5$, it indicates that b dominates a . If $P = 0.5$, they are non-dominated [25].

2.3.2. IDM2 based on conditional interval confidence

$$P(a \leq b) = \begin{cases} \frac{d(b, \beta) - d(a, b)}{d(a, \beta) + d(b, \beta)}, & \text{if } m(a) = m(b) \wedge \underline{a} - \underline{b} \geq w(a) \\ \frac{d(b, \beta)}{d(a, \beta) + d(b, \beta)}, & \text{otherwise} \end{cases} \quad (7)$$

This method is an improvement of 2.3.1. When the midpoints of intervals a and b are the same, the difference between the lower bounds of a and b is bigger than the interval width of a . This indicates that a has better convergence potential, so a dominates b [45].

2.3.3. IDM3 based on interval confidence level

$$P(a \leq b) \stackrel{\Delta}{=} \max \left\{ 1 - \max \left[\frac{\bar{a} - \underline{b}}{(\bar{a} - \underline{a}) + (\bar{b} - \underline{b})}, 0 \right], 0 \right\} \quad (8)$$

According to this method, the interval confidence level P is calculated. If $P > \gamma$, it indicates that a dominates b ($\gamma \in [1/2, 1]$). If $P < 1 - \gamma$, it indicates that b dominates a . If $P \in [1 - \gamma, \gamma]$, it means they are non-dominated [44].

2.3.4. IDM4 based on improved interval confidence level

$$\begin{aligned} P(a \leq b) &\stackrel{\Delta}{=} \max \left\{ 1 - \max \left[\frac{\bar{a} - \underline{b} - \Delta P(a, b)}{(\bar{a} - \underline{a}) + (\bar{b} - \underline{b})}, 0 \right], 0 \right\} \\ \Delta P(a, b) &= (m(b) - m(a)) - (\underline{a} - \underline{b}) \end{aligned} \quad (9)$$

This method is an improvement upon 2.3.3. It emphasizes the midpoint and lower bound of the interval and introduces the relative difference $\Delta P(a, b)$ to select intervals with either a better average convergence level or better convergence potential [42].

2.4. Interval pareto dominance

The Pareto dominance between two solutions can be obtained by using IDM.

$$\begin{aligned} \mathbf{x} < \mathbf{y} : & \forall i \in M, f_i(\mathbf{x}, c_i) \leq f_i(\mathbf{y}, c_i) \\ & \wedge \exists j \in M, f_j(\mathbf{x}, c_i) < f_j(\mathbf{y}, c_i) \end{aligned} \quad (10)$$

$\mathbf{x} < \mathbf{y}$ represents that x dominates y . $f_i(\mathbf{x}, c_i) \leq f_i(\mathbf{y}, c_i)$ represents that, after using IDM, the i th objective interval of solution x dominates the i th objective interval of solution y or they are non-dominated. $f_j(\mathbf{x}, c_i) < f_j(\mathbf{y}, c_i)$ represents the i th objective interval of solution x dominates the i th objective interval of solution y .

2.5. Information entropy

“Entropy” was initially introduced as a concept in thermodynamics and later adopted in information theory to measure the degree of disorder in a system [47]. After its establishment in the field of information theory, it has been applied in decision-making involving multiple evaluation attributes [51–53].

Attribute entropy reflects the stability of the overall data in a certain attribute. Assuming there are N individuals and M evaluation attributes in a decision-making scenario, the entropy calculation for all individuals in the k th $k \in (1, 2, \dots, M)$ evaluation attribute is as follows:

$$H_k = -\frac{1}{\ln N} \sum_{j=1}^N p_{kj} \ln p_{kj} \quad (11)$$

$$p_{kj} = \frac{f_k(\mathbf{x}_j)}{\sum_{j=1}^N f_k(\mathbf{x}_j)} \quad (12)$$

p_{kj} represents the proportion of the k th attribute value of the j th individual in the overall individuals. $f_k(\mathbf{x}_j)$ represents the normalized value of the k th attribute for the j th individual. H_k represents the entropy of the k th attribute.

When all N individuals have the same value for the k th attribute, $H_k = 1$, indicating individuals of this attribute are completely stable. Therefore, the larger the value, the less useful information the k th attribute provides to the decision-maker. The smaller the value, the more crucial the evaluation attribute is considered to be.

2.6. Interval crowding distance

In interval optimization algorithms, individuals are represented as cubes in the objective space. The formula for calculating the crowding distance of an individual is as follows [26]:

$$C(\mathbf{x}_1) = \frac{D(\mathbf{x}_1, \mathbf{x}_2) + D(\mathbf{x}_1, \mathbf{x}_3)}{2} \quad (13)$$

$$D(\mathbf{x}_1, \mathbf{x}_2) = \frac{\sum_{g=1}^M |m(f_g(\mathbf{x}_1, c)) - m(f_g(\mathbf{x}_2, c))|}{v(\mathbf{x}_1, \mathbf{x}_2) + V(\mathbf{x}_1) + V(\mathbf{x}_2) + 1} \quad (14)$$

$D(\mathbf{x}_1, \mathbf{x}_2)$ represents the distance from $\mathbf{x}_1$ to $\mathbf{x}_2$. $\mathbf{x}_2$ and $\mathbf{x}_3$ are the two solutions closest to $\mathbf{x}_1$. $m(f_g(\mathbf{x}_1, c))$ represents the midpoint of the g th interval objective, $v(\mathbf{x}_1, \mathbf{x}_2)$ is the overlap area between $\mathbf{x}_1$ and $\mathbf{x}_2$, and $V(\mathbf{x}_1)$ represents the volume of $\mathbf{x}_1$ in the objective space.

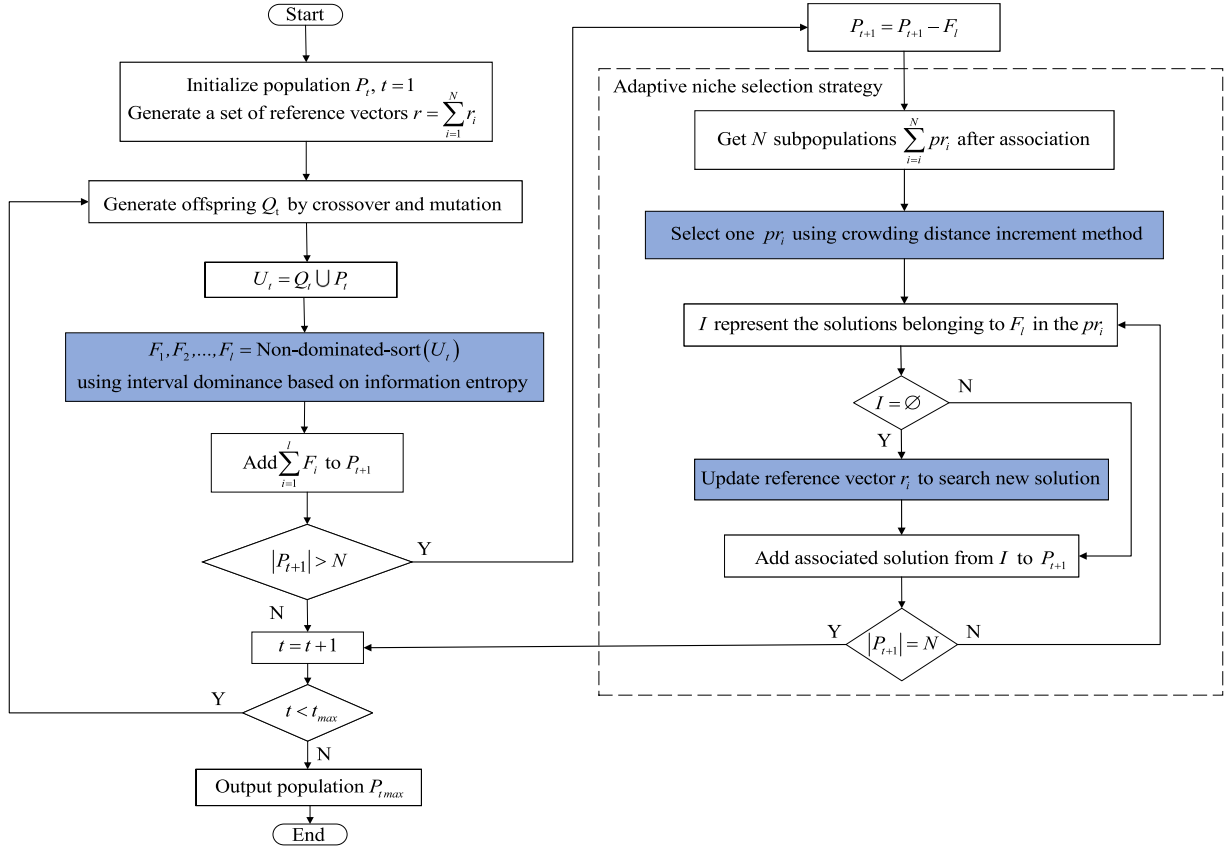


Fig. 1. Flowchart of IMEA-IED.

3. The proposed IMEA-IED

This section will provide a detailed introduction to the proposed

Algorithm 1: IMEA-IED

Input: population size N , objective number M , population P_0 and uniformly distributed reference vectors
 $r = (r_1, r_2, \dots, r_N)$, current generation t , max generations t_{max}

Output: Last generation population $P_{t_{max}}$

```

1  $t = 0$ ;
2 repeat
3    $i = 0$ ;
4    $Q_{t+1} = \text{Apply binary crossover and polynomial mutation}(P_t)$ ;
5    $U_{t+1} = P_t \cup Q_{t+1}$ ;
6    $(F_1, F_2, \dots, F_t) = \text{Non-dominated-sort}(U_{t+1})$  using interval dominance based on information entropy;
7   while  $|P_{t+1}| < N$  do
8      $i = i + 1$ ;
9      $P_{t+1} = P_{t+1} \cup F_i$ ;
10  end
11  if  $|P_{t+1}| > N$  then
12     $P_{t+1} = P_{t+1} - F_l$ ;
13    last front  $F_l = F_i$ ;
14     $K = |P_{t+1}| - N$ ;
15    Selecting  $K$  solution from  $F_l$  using adaptive niche selection strategy to add to  $P_{t+1}$ ;
16  end
17   $t = t + 1$ ;
18 until  $t \geq t_{max}$ ;

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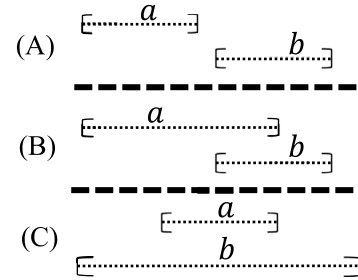


Fig. 2. Three types of interval relationships.

IMEA-IED. Section 3.1 introduces the overall framework, Section 3.2 presents the proposed interval dominance method based on information entropy, and Section 3.3 introduces the adaptive niche selection strategy.

3.1. The framework of IMEA-IED

The pseudo-code of the algorithm is shown as Algorithm 1. First, the algorithm initializes the population P_0 and generates a set of uniformly distributed reference vectors r . The parent generates offspring through crossover and mutation operators (Line 4). The combined population is subjected to non-dominated sorting and ranking using the proposed interval dominance method (Lines 5–6). Solutions are selected from front [54] as parent solutions for the next generation P_{t+1} (Line 7–10). If the number of parent solutions exceeds N after the above operations, an adaptive niche selection strategy is employed to select K solutions from the last non-dominated front F_l (Line 11–16), l is last front index.

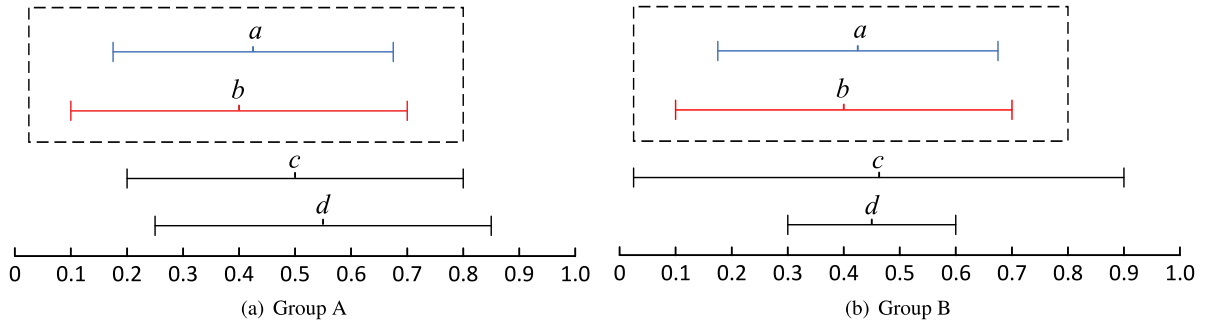


Fig. 3. Two interval groups with different overall relationships.

The algorithm flowchart is shown in Fig. 1. Colored boxes indicate a proposed method involved in the step. The left-colored box indicates that the interval dominance method based on information entropy is used in non-dominance sorting. In the large box on the right is the proposed adaptive niche selection strategy, where the first color box uses the crowding increment method to select subpopulations, and the second color box uses the update reference vector method to explore new associated solutions.

3.2. Interval dominance method based on information entropy

The non-dominated sorting needs to compare the interval objective values of different individuals. The position relationship between intervals is shown in Fig. 2. In Case A, any value of a is less than b . It is easy to infer that a dominates b from the convergence perspective. When in the crossover relationship (Case B) or the contain relationship (Case C), we can not only compare the intervals from the convergence perspective. In both cases, we need to consider both convergence and uncertainty.

As shown in Fig. 3, groups A and B have the same two intervals $a = [0.18, 0.68]$, $b = [0.1, 0.7]$. The width of a is smaller than b , and the midpoint of b is smaller than a . The midpoint measures the average convergence degree of the interval, while the width measure measures the uncertainty of the interval [45]. Therefore, the smaller the width and midpoint of the interval, the better. IDM1 and IDM2 in Section 2.3 calculate that a dominates b . In contrast, IDM3 and IDM4 calculate that b dominates a ($\gamma = 0.5$). Whenever the interval relationship mentioned above occurs during the evolutionary population, IDM1 and IDM2 will always choose a , which will miss some solutions with good convergence. IDM3 and IDM4 will always choose b , which will miss some solutions with good uncertainty. However, when observing the overall interval relationship, the difference in the interval width of group A is small, while the midpoint difference of the intervals is large. The midpoint difference of group B is small, while the width difference is large. Therefore, comparing intervals in group A should focus on one with good convergence, while group B needs to focus on one with good uncertainty.

After the above analysis, it is necessary to consider the overall interval relationship when comparing intervals, and the dominance method should adaptively select intervals. So an interval dominance method based on information entropy is proposed. Pseudocode as shown in Algorithm 2. Before comparing two individuals x_i and x_j in the k th objective interval, firstly calculate the convergence entropy and uncertainty entropy of the entire population at this objective.

$$H_{ck} = -\frac{1}{\ln N} \sum_{j=1}^N p_{ckj} \ln p_{ckj} \quad (15)$$

$$p_{ckj} = \frac{m(f_k(x_j))}{\sum_{j=1}^N m(f_k(x_j))} \quad (16)$$

$$H_{wk} = -\frac{1}{\ln N} \sum_{j=1}^N p_{wkj} \ln p_{wkj} \quad (17)$$

Algorithm 2: Interval Dominance Method based on Information Entropy

Input: population size N , objective number M , population P_0

Output: sorted fronts $(F_1, F_2, \dots, F_I)$

```

1 for  $i=1:N$  do
2   for  $j=1:N$  do
3     for  $k=1:M$  do
4        $a = f_k(x_i)$ ,  $b = f_k(x_j)$ ;
5       if  $H_{ck} < H_{wk}$  then
6          $P(a, b) = P_c(a \leq b)$ ;
7       else
8         if  $H_{ck} > H_{wk}$  then
9            $P(a, b) = P_w(a \leq b)$ ;
10        end
11        if  $H_{ck} == H_{wk}$  then
12          Choose the traditional IDM3 to compare  $a, b$ ;
13        end
14      end
15    end
16  end
17 end
18 Obtain sorted fronts by Non-dominated-sorting;

```

$$p_{wkj} = \frac{w(f_k(x_j))}{\sum_{j=1}^N w(f_k(x_j))} \quad (18)$$

H_{ck} represents the convergence entropy of the k th objective, N represents the population size, and $m(f_k(x_j))$ represents the midpoint of the j th individual in the k th objective interval. p_{ckj} represents the proportion of $m(f_k(x_j))$ to the overall population. H_{wk} represents the uncertainty entropy of the k th objective, and $w(f_k(x_j))$ represents the width of the j th individual in the k th objective interval.

Section 2.4 mentions that information entropy can measure the stability of all individuals on a certain attribute. If $H_{ck} < H_{wk}$, it indicates that the average convergence level of the k th objective is more unstable, and interval comparison should focus on convergence. As previously mentioned, the midpoint represents the mean convergence level. Therefore, the convergence-based interval confidence level is proposed:

$$P_c(a \leq b) \stackrel{\Delta}{=} \max \left\{ 1 - \max \left[\frac{\bar{a} - \underline{b} - \Delta m}{(\bar{a} - \underline{a}) + (\bar{b} - \underline{b})}, 0 \right], 0 \right\} \quad (19)$$

$$= \max \left\{ 1 - \max \left[\frac{1}{2} + \frac{m(a) - m(b) - \Delta m}{w(a) + w(b)}, 0 \right], 0 \right\}$$

Eq. (19) introduces a midpoint difference Δm based on IDM3, $\Delta m = m(b) - m(a)$. For a and b in Fig. 3, the confidence level $P_c(b \leq a)$ is larger

than the $P(b \leq a)$ of IDM3. Thus the probability of a midpoint optimal interval dominating other intervals increases.

If $H_{ck} > H_{wk}$, it indicates that the width of the k th objective is more unstable, and interval comparison should focus on uncertainty. The width represents the interval uncertainty. Therefore, the uncertainty-based interval confidence level is proposed:

$$P_w(a \leq b) = \max \left\{ 1 - \max \left[\frac{\bar{a} - \underline{b} - \Delta w}{(\bar{a} - \underline{a}) + (\bar{b} - \underline{b})}, 0 \right], 0 \right\} \quad (20)$$

$$= \max \left\{ 1 - \max \left[\frac{1}{2} + \frac{m(a) - m(b) - \Delta w}{w(a) + w(b)}, 0 \right], 0 \right\}$$

Eq. (20) introduces a width difference Δw based on IDM3, $\Delta w = w(b) - w(a)$. The probability of a width optimal interval dominating other intervals increases.

Next, we apply the proposed information entropy dominance method to compare a and b in Fig. 3. The four intervals of population A are [0.18, 0.68], [0.1, 0.7], [0.02, 0.9], [0.3, 0.6], and the calculated H_c is 0.765, H_w is 0.722. $H_w < H_c$, Eq. (20) can obtain that a dominates b . The four intervals of population B are [0.18, 0.68], [0.1, 0.7], [0.2, 0.8], [0.25, 0.85], and the calculated H_c is 0.679, H_w is 0.793. $H_c < H_w$, Eq. (19) can obtain that b dominates a . Therefore, the proposed method can adaptively select intervals more conducive to population evolution, thereby balancing convergence and uncertainty.

3.3. Adaptive niche selection strategy

The proposed adaptive niche selection strategy aims to adapt the niche strategy to IMAOPs and thus we construct the association between interval solutions and reference vectors. In addition, considering the need to improve the ability of niche count to select subpopulations, the crowding increment method is proposed to enhance the selection capability. Considering this strategy's lack of update processing for empty subpopulations, an adaptive update reference vector method for empty subpopulations is proposed to explore solutions for new regions. The specific process can be found in Algorithm 3.

3.3.1. Association method

At first, the association method calculates the perpendicular distance from the solutions of P_{t+1} and F_l to the reference vectors $r_1, r_2, \dots, r_N$, as shown in formula (22). Each solution is associated with the corresponding reference vector that has the smallest perpendicular distance d_2 . The associated solutions of one reference vector constitute a subpopulation, thus N subpopulations correspond to N vectors, represented as $pr_1, pr_2, \dots, pr_N$. In addition, there may be multiple solutions associated with one reference vector, resulting in some subpopulations being empty.

$$d_1 = \frac{\|f(\mathbf{x}, \mathbf{c})^T r\|}{\|r\|} \quad (21)$$

$$d_2 = \|f(\mathbf{x}, \mathbf{c}) - d_1 r\| \quad (22)$$

$f(\mathbf{x}, \mathbf{c})$ represents a solution vector in the objective space, while r represents one of the reference vectors. Based on the interval arithmetic operations mentioned in section 2.2, it is not difficult to calculate d_1 and d_2 . d_1 represents the length of the projection of the solution vector in the objective space onto the vector r , while d_2 represents the perpendicular distance from the solution vector to the reference vector r .

Algorithm 3: Adaptive Niche Selection Strategy

Input: selected population P_{t+1} , last front F_l , reference vectors $r_1, r_2, \dots, r_N$, subpopulation $pr_1, pr_2, \dots, pr_N$, sets of associated solutions in P_{t+1} for N reference vectors $R_1, R_2, \dots, R_N$, crowding distance increment of each subpopulation $CI(i)$, index of the vector associated with each solution $\rho(x)$

Output: selected population P_{t+1}

```

1  flag = 0;
2  while  $|P_{t+1}| < N$  do
3     $J = \{i : \argmin(|R_i|)\}$ ;
4    if  $|J| > 1$  then
5       $J = \argmax_{i \in J} (CI(i))$ ;
6    else
7       $J = J$ ;
8    end
9     $I_J = \{x : \rho(x) = r_J, x \in F_l\}$ ;
10   if  $I_J \neq \emptyset$  then
11     if  $|R_J| = 0$  then
12        $P_{t+1} = P_{t+1} \cup \{x_k = \argmin_{x \in I_J} (d_2(x, r_J))\}$ ;
13        $R_J = R_J \cup x_k$ ;
14        $F_l = F_l \cup x_k$ ;
15     else
16       select one  $x_k$  randomly from  $I_J$  in  $F_l$  to enter  $P_{t+1}$ ;
17        $R_J = R_J \cup x_k$ ;
18        $F_l = F_l \cup x_k$ ;
19     end
20   else
21     if flag=0 then
22       update reference vector get  $I_J$ ;
23       %  $I_J$ : Solution associated with the new reference vector;
24     end
25     if  $I_J \neq \emptyset$  then
26        $P_{t+1} = P_{t+1} \cup I_J$ ;
27        $R_J = R_J \cup I_J$ ;
28        $F_l = F_l \cup I_J$ ;
29       flag = 1;
30     else
31       exclude  $r_J$  for the current generation;
32     end
33   end
34 end
```

3.3.2. The crowding distance increment method

According to the previous context, we obtained N subpopulations. Subsequently, one subpopulation is selected through the comparison of niche counts [29]. However, many subpopulations have the same niche count, leading to a decrease in the selection ability of this method. In such cases, the proposed method of crowding distance increment is used for comparison.

$$CI(i) = C_i(i) - C_{i-1}(i) \quad (23)$$

$$J = \{i : \argmax(CI(i))\} \quad (24)$$

$C_i(i)$ represents the crowding distance sum of the i th subpopulation during the i th iteration, and $CI(i)$ represents its crowding distance increment. J represents the index of the subpopulation with the maximum crowding distance increment.

A larger value of CI indicates a more robust evolutionary capability of the subpopulation diversity, while a smaller value suggests a slower evolution or reaching a saturation state. Therefore, it can be assumed that the subpopulation with the maximum CI exhibits a more prominent performance, and the J th subpopulation is selected (Lines 4–6).

3.3.3. Adaptive update reference vectors method

After the above operations, a subpopulation pr_J is selected and r_J is its corresponding reference vector. Then an associated solution

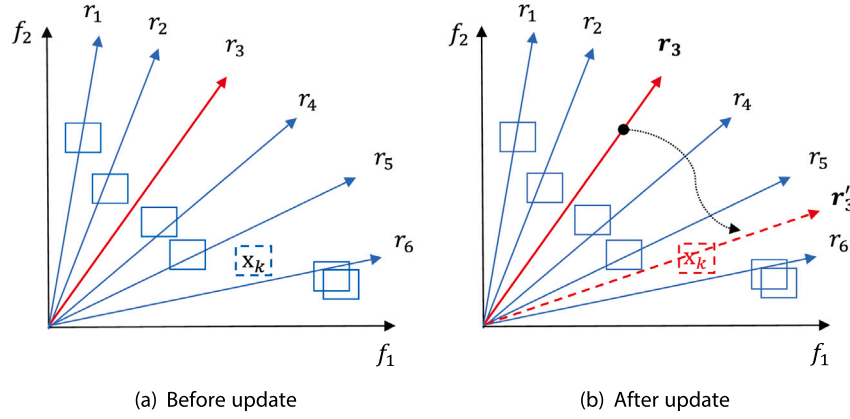


Fig. 4. The correlation between a set of reference vectors and the last front solution.

will be selected. First, find all associated solutions of r_j in F_l : $I_j = \{x : \rho(x) = r_j, x \in F_l\}$, $\rho(x)$ represents the reference vector associated with solution x (Line 9).

If $I_j \neq \emptyset$, then check if $|R_j|$ (the niche count of r_j) is equal to 0. If $|R_j| = 0$, select the solution closest to r_j in F_l to enter population P_{t+1} . Otherwise, select the solution randomly from the subpopulation in F_l to enter population P_{t+1} (Line 10–19).

If $I_j = \emptyset$, this indicates that it is not currently feasible to select a solution from this subpopulation. So the position of the reference vector r_j is temporarily updated to find a solution using the following method:

$$r_{\text{new}} = (r_k + r_l) / 2 \quad (25)$$

Suppose x_k is the solution with the maximum crowding distance in F_l , and r_k is its associated reference vector. r_l is a reference vector that is the second closest to x_k in terms of d_2 . r_{new} is generated between them. Then calculate whether there are any solutions in F_l associated with r_{new} . If there are, select the solution closest to r_{new} to enter population P_{t+1} . If not, the reference r_j is excluded from further consideration for the current generation (Line 20–33).

Fig. 4 provides a more intuitive representation of the method. The figure shows r_1 to r_6 reference vectors and their respective associated solutions in F_l . The associated vector for the solution x_k with the maximum crowding distance is r_5 . Assuming that the r_3 and its subpopulation are selected at this time and it has no associated solution in F_l , then the position of r_3 is temporarily updated to r'_3 using Eq. (25). At this point, x_k is closest to r'_3 , select it to enter P_{t+1} . According to the traditional niche selection strategy, r_5 would choose the closest solution to enter P_{t+1} , which means x_k has a certain probability of not being selected. Therefore, our method helps to select more solutions that can enhance diversity. It should be noted that the newly generated vectors by this method only serve as an auxiliary selection in the current situation and will not be passed on to the next generation, and the adaptive updating of reference vectors method is executed once per generation.

3.4. Computational complexity

In the proposed IMEA-IED, the complexity for calculating the interval information entropy is $O(2mN)$, where m is the number of objectives and N is the population size. The complexity for performing dominance comparisons is $O(mN^2)$. In the niche selection, the complexity for population partitioning is $O(mNH)$, where H represents the number of reference vectors. The complexity for calculating the sum of crowding distances for each subpopulation is $O(Hk)$, where k represents the number of solutions in a subpopulation. so the complexity for crowding distance increment for each subpopulation is $O(H)$. The reference vector update operation is performed only once in each iteration.

4. Experiment

The experiment mainly focuses on algorithm evaluation. It compares our algorithm with seven interval optimization algorithms, namely IMOEA [25], SetGA [35], DI- μ MOGA [39], InMaOEA [26], IMOEA/D [12], EG-IMOEA [45] and IMOEA/D-AWN [55] on four performance metrics using 60 IMAOPs and a practice problem. The goal is to demonstrate the effectiveness of our algorithm. All algorithms in the experiment were run on a machine with a configuration of Windows 10 operating system, Intel (R) Core (TM) i9-13900k CPU @ 3.0 GHz. The experiment was conducted using MATLAB R2021b.

4.1. Benchmark test problems

The selected problems are WFG1–9 [49] and DTLZ1–6 [8], which have a scalable number of objectives. For different numbers of objectives, the number of variables in these problems is defined as $n = k + l$, where k and l are referred to as position and distance-related parameters, respectively. For WFG1–WFG9, $k = m - 1$ and $l = 20$. For DTLZ1, $l = 5$; for DTLZ2–6, $l = 10$. The above-mentioned test problems can be transformed into interval optimization problems by adding a different perturbation factor c to each objective value [35]. The perturbation factors used in this study are presented as follows.

$$c_i = \lambda \left| \sin \left(10i\pi \sum_j x_j \right) \right| f_i(x) \quad (26)$$

$$f_i(x, c_i) = [f_i(x) - c_i, f_i(x) + c_i] \quad (27)$$

c_i represents the perturbation factor for the i th objective. The value of λ is set to 0.1. $f_i(x)$ represents the value of the i th objective function for the solution x in the deterministic optimization problem, and x_j represents the j th decision variable of x . $f_i(x, c_i)$ represents the transformed interval objective.

By applying this operation to the WFG and DTLZ problems, we obtain the interval optimization problems, denoted as InWFG and InDTLZ. InWFG and InDTLZ have the same characteristics as WFG and DTLZ, respectively.

4.2. Performance metrics

In the experiment, four metrics were used for performance comparison.

(1) Interval Inverse Generation Distance (IIGD)

The IIGD metric is used to measure the convergence and diversity of algorithms [55]. Using the Pareto optimal front of each deterministic benchmark test problem as the true Pareto front for each uncertain problem, the IIGD is the average distance between the interval mid-points of the optimal solution set obtained by the interval optimization

Table 1

Population size of different objective numbers.

Number of objectives	Popsiz
5	212 (210)
8	156 (156)
10	276 (275)
15	136 (135)

Table 2Wilcoxon rank-sum test between IMEA-IED ($\gamma = 0.6$), IMEA-IED ($\gamma = 0.7$), IMEA-IED ($\gamma = 0.8$), IMEA-IED ($\gamma = 0.9$) and IMEA-IED ($\gamma = 0.5$).

$\gamma = 0.5$ v.s.		$\gamma = 0.6$	$\gamma = 0.7$	$\gamma = 0.8$	$\gamma = 0.9$
IIGD	+	8/60	9/60	17/60	23/60
	-	1/60	0/60	0/60	3/60
	=	51/60	51/60	43/60	34/60
Imprecision	+	10/60	14/60	18/60	19/60
	-	0/60	0/60	1/60	0/60
	=	50/60	46/60	41/60	41/60

algorithm and the true Pareto front. A smaller IIGD value indicates better performance.

(2) Spread

The smaller the spread value, the better the distribution of the approximate optimal solution set [49].

(3) Imprecision

In the approximate optimal solution set, the perimeter of all solutions formed by the cube is called the Imprecision. Smaller values of Imprecision are better [35,56]. This section compares the average Imprecision value of the optimal solution set.

(4) Runtime

Runtime is used to evaluate the execution time of algorithms.

4.3. Experimental settings

For all test problems, the population size is set according to Table 1 [29]. The number of iterations is set to 1000. Within the parenthesis is the setting for IMOE/D and IMOE/D-AWN, and outside the parenthesis is the setting for other algorithms. In order to achieve fair comparison as much as possible, the parameters for all comparative algorithms are set according to their original research papers. The number of solutions for each set in SetGA is 4. The interval-valued Chebyshev scalar function was used in the IMOE/D, the neighborhood size was set to 10, and the number of reference vectors was set to the population size. The start generation for adjusting the reference vectors was set to half of the maximum generation G_{max} , and the adjustment period was set to $G_{max}/6$, rounded down to the nearest integer. The ratio of generations to evolve of IMOE/D-AWN is set to 0.8, the adjustment period is set to 100, the ratio of the updated weight vector is set to 0.05, and $nus = 0.05 * N$. The γ value of the interval confidence level in IMEA-IED is set to 0.5. The crossover and mutation probabilities of IMOE/D, SetGA, InMaOE/D, EG-IMOE/D, IMOE/D-AWN, and IMEA-IED are set to 0.9 and 0.1, respectively. And the crossover and mutation probabilities of DI- μ MOGA are set to 0.8 and 0.05, respectively. Additionally, the algorithms are run for a total of 20 independent runs.

4.4. Experimental results and analysis

4.4.1. Effects of different values of γ

In the proposed IMEA-IED, there exists a parameter γ , as shown in Section 2.3.3, where $\gamma \in [1/2, 1]$. The parameter γ represents the threshold for confidence level in the dominance methods, with larger values of γ indicating more stringent conditions for one interval to dominate another. However, previous studies have rarely discussed the impact of different γ -values on algorithm performance.

This section employs Wilcoxon rank-sum test [57] to examine the significant differences of IMEA-IED with different values of γ compared to IMEA-IED ($\gamma = 0.5$). The labels “+”, “-”, and “=” denote that the results of the other algorithms are significantly better, worse, or equal to the results of IMEA-IED ($\gamma = 0.5$), respectively. Table 2 shows that the performance of algorithms with other values of γ is better to varying degrees than IMEA-IED ($\gamma = 0.5$). Out of 60 test problems, IMEA-IED ($\gamma = 0.9$) shows significantly better IIGD performance than IMEA-IED ($\gamma = 0.5$) in 23 cases. IMEA-IED ($\gamma = 0.9$) also shows significantly better Imprecision performance than IMEA-IED ($\gamma = 0.5$) in 19 cases. It can be seen that IMEA-IED ($\gamma = 0.9$) outperforms IMEA-IED ($\gamma = 0.5$) most times among algorithms with different parameter values. Therefore, the algorithm performs best when γ is set to 0.9.

4.4.2. Effectiveness of algorithm

This section compares IMEA-IED with 7 existing algorithms to demonstrate the algorithm's effectiveness. Table 3, 5, 7 shows the average and standard deviation of IIGD, Imprecision, and Spread obtained by different algorithms on InWFG. Table 4, 6, 8 shows the average and standard deviation of IIGD, Imprecision, and Spread obtained by different algorithms on InDTLZ. The best results are highlighted with the gray background, and the Wilcoxon rank-sum test is used to evaluate the superiority of IMEA-IED compared to other algorithms. The labels “+”, “-”, and “=” indicate that the results of the other algorithms are significantly better, worse, or equal to the results of IMEA-IED, respectively.

Tables 3–4 shows the comparison of 8 algorithms in IIGD indicators. IIGD can comprehensively measure the convergence and diversity of interval algorithms. In InWFG4-7 and InDTLZ1-4, IMEA-IED's IIGD indicators are superior to all the compared algorithms. The Pareto front of InWFG4-9 and InDTLZ2-4 are concave. The Pareto front of InDTLZ1 is linear. This indicates that in these concave-type and linear-type problems, the proposed algorithm utilizes reference vectors to achieve good distribution in the objective space. The crowding distance increment (CDI) method and the adaptive update reference vectors (AUV) method can improve population diversity. In addition, the problems of non-uniform and concave lead to large differences in convergence between intervals, so the information entropy dominance method can adaptively improve population convergence. InWFG1 features a sharp tail front, in which the proposed algorithm is superior to other algorithms in 5 and 8 objectives. However, IMEA-IED is inferior to the decomposition-based IMOE/D and IMOE/D-AWN in 15 objectives. It may be that decomposition-based methods [58] are suitable for solving sharp-tail problems with more objectives. In the convex and disconnected InWFG2, the proposed algorithm outperforms other algorithms on 5 and 8 objectives, while IMOE/D-AWN performs best on 10 and 15 objective problems. The possible reason is that the increase in the number of objectives leads to a weaker selection ability of the non-dominated sorting, resulting in IMEA-IED being inferior to IMOE/D-AWN. The IMEA-IED is superior to IMOE/D, SetGA, DI- μ MOGA, and InMaOE/D in InWFG3, but inferior to EG-IMOE/D, IMOE/D, and IMOE/D-AWN. This is because the degenerate problem leads to many reference vectors not being utilized, thus preventing the adaptive niche selection strategy from achieving its full impact. The superior performance of IMOE/D on InDTLZ5-6 indicates that the decomposition approach and neighborhood crossover contribute to the convergence of solutions towards the degenerate Pareto Front [59].

Tables 5–6 shows the comparison of eight algorithms in Imprecision indicators. The metric Imprecision can measure the uncertainty of algorithms. The proposed algorithm is superior to IMOE/D, InMaOE/D, and IMOE/D in all problems. It is also superior to all algorithms in the uncertainty of InWFG2-InWFG8. This proves that the information entropy dominance method of the proposed algorithm can achieve good uncertainty. It can be observed that SetGA exhibits the least uncertainty on InWFG1 and InDTLZ5-6. This is likely attributable to that Imprecision

Table 3

The mean and standard deviation of IIGD obtained by different algorithms on InWFG.

Problem	M	IMOEa	SetGA	DI- μ MOGA	InMaOEA	IMOEa/D	EG-IMOEa	IMOEa/D-AWN	IMEA-IED
InWFG1	5	1.3633e+0 (3.19e-1)	1.2164e+0 (4.23e-2)	2.2438e+0 (8.49e-1)	1.2411e+0 (2.32e-1)	9.9237e-1 (1.96e-3)	6.8342e-1 (1.56e-1)	3.1199e+0 (2.75e+0)	5.9806e-1 (1.36e-2)
	8	2.0238e+0 (2.76e-1)	2.0601e+0 (7.70e-2)	2.8812e+0 (2.50e-1)	1.8234e+0 (2.19e-1)	1.4166e+0 (5.72e-2)	1.6752e+0 (3.23e-1)	1.6117e+0 (9.98e-2)	1.3605e+0 (5.71e-2)
	10	2.0994e+0 (2.11e-1)	2.0233e+0 (1.03e-1)	3.1739e+0 (4.47e-2)	2.4182e+0 (9.24e-1)	1.4896e+0 (7.80e-2)	1.7725e+0 (1.46e-1)	1.5710e+0 (1.25e-1)	1.5210e+0 (5.53e-2)
	15	6.1313e+0 (5.64e+0)	2.7363e+0 (7.41e-2)	4.1720e+0 (2.20e-1)	5.1594e+0 (3.34e+0)	2.1487e+0 (4.52e-2)	3.0932e+0 (7.49e-1)	2.4302e+0 (1.47e-1)	2.6780e+0 (2.63e-1)
InWFG2	5	1.9080e+0 (3.38e-1)	1.4825e+0 (1.51e-1)	1.9061e+0 (1.62e-1)	1.9005e+0 (2.19e-1)	1.3788e+0 (5.06e-2)	7.2451e-1 (4.54e-2)	7.6920e-1 (8.44e-2)	6.1291e-1 (2.24e-2)
	8	4.8027e+0 (4.07e-1)	2.4559e+0 (2.77e-1)	5.5863e+0 (7.41e-1)	1.9005e+0 (2.19e-1)	2.0894e+0 (3.74e-2)	1.8738e+0 (2.26e-1)	1.7777e+0 (1.48e-1)	1.6061e+0 (1.12e-1)
	10	6.8451e+0 (1.34e+0)	2.4421e+0 (4.95e-1)	1.0475e+1 (2.10e+0)	6.6996e+0 (2.34e+0)	1.9097e+0 (1.05e-1)	2.1003e+0 (3.30e-1)	1.4969e+0 (9.81e-2)	1.6830e+0 (1.11e-1)
	15	1.5739e+1 (7.61e-1)	6.2590e+0 (4.49e+0)	1.5012e+1 (2.81e+0)	1.6844e+1 (1.77e+0)	2.5661e+0 (8.90e-2)	7.0787e+0 (2.21e+0)	2.1342e+0 (1.41e-1)	2.3407e+0 (1.03e-1)
InWFG3	5	2.3799e+0 (2.29e-1)	4.5955e+0 (9.92e-1)	2.8154e+0 (5.63e-2)	2.3700e+0 (1.68e-1)	1.4718e+0 (2.12e-2)	1.9992e-1 (4.61e-2)	1.1949e+0 (2.63e-1)	1.1371e+0 (4.44e-2)
	8	5.1653e+0 (3.24e-1)	8.1751e+0 (6.39e-1)	5.1732e+0 (4.77e-1)	5.2472e+0 (2.72e-1)	2.4459e+0 (1.73e-1)	5.5868e-1 (9.45e-2)	2.0747e+0 (3.90e-1)	4.1286e+0 (7.93e-1)
	10	6.5330e+0 (3.46e-1)	9.6571e+0 (1.54e+0)	7.1928e+0 (5.21e-1)	6.5153e+0 (2.40e-1)	3.2761e+0 (9.27e-2)	6.5415e-1 (1.33e-1)	2.4567e+0 (3.90e-1)	5.4623e+0 (9.62e-1)
	15	7.1312e+0 (5.32e-1)	1.6420e+1 (3.24e+0)	1.7312e+1 (1.03e+0)	9.7761e+0 (4.00e-1)	4.6029e+0 (2.59e-1)	2.1730e+0 (4.43e-1)	4.5220e+0 (5.28e-1)	6.6852e+0 (1.45e+0)
InWFG4	5	2.0658e+0 (1.46e-1)	3.5873e+0 (1.43e+0)	6.5919e+0 (4.11e-1)	2.1596e+0 (1.47e-1)	2.1513e+0 (7.05e-3)	1.4249e+0 (9.54e-2)	1.6010e+0 (1.18e-1)	1.2506e+0 (1.16e-2)
	8	4.7908e+0 (2.13e-1)	1.0182e+1 (2.27e+0)	1.2315e+1 (2.52e-1)	4.6397e+0 (2.14e-1)	5.3117e+0 (3.77e-1)	6.5402e+0 (7.57e-1)	4.6552e+0 (2.30e-1)	3.8581e+0 (5.87e-2)
	10	6.4433e+0 (2.48e-1)	1.2697e+1 (2.39e+0)	1.6279e+1 (8.44e-1)	6.3285e+0 (2.29e-1)	6.8264e+0 (4.15e-1)	9.2261e+0 (1.20e+0)	5.7909e+0 (2.51e-1)	5.6401e+0 (1.14e-1)
	15	1.2832e+1 (5.65e-1)	2.5808e+1 (1.72e+0)	2.7994e+1 (2.94e-1)	1.2631e+1 (4.42e-1)	1.3867e+1 (1.42e+0)	2.0231e+1 (1.35e+0)	1.1295e+1 (9.81e-1)	1.0645e+1 (2.17e-1)
InWFG5	5	2.1019e+0 (1.70e-1)	3.0009e+0 (1.28e+0)	7.2212e+0 (2.18e-1)	1.9609e+0 (6.41e-1)	2.8482e+0 (8.30e-2)	1.6242e+0 (7.61e-2)	1.9665e+0 (1.20e-1)	1.2445e+0 (1.06e-2)
	8	5.0224e+0 (2.77e-1)	8.0001e+0 (1.27e+0)	1.3285e+1 (2.76e-1)	4.8976e+0 (2.45e-1)	5.1060e+0 (1.31e-1)	5.1025e+0 (4.90e-1)	4.5236e+0 (1.71e-1)	4.0068e+0 (9.06e-2)
	10	6.8518e+0 (3.42e-1)	9.1548e+0 (2.11e+0)	1.6014e+1 (9.79e-1)	6.7586e+0 (2.96e-1)	7.0789e+0 (1.97e-1)	6.9934e+0 (9.47e-1)	5.9429e+0 (2.15e-1)	5.7116e+0 (1.09e-1)
	15	1.3127e+1 (5.98e-1)	2.5983e+1 (2.42e+0)	2.8527e+1 (1.09e+0)	1.3226e+1 (5.52e-1)	1.2698e+1 (4.79e-1)	1.6003e+1 (8.78e-1)	1.1745e+1 (8.01e-1)	1.0613e+1 (1.59e-1)
InWFG6	5	2.6060e+0 (1.12e-1)	4.6562e+0 (1.41e+0)	6.9384e+0 (3.31e-1)	2.5755e+0 (2.00e-1)	2.9053e+0 (6.55e-2)	1.7525e+0 (7.38e-2)	2.2497e+0 (2.80e-1)	1.2430e+0 (1.73e-2)
	8	5.0676e+0 (1.35e-1)	9.8928e+0 (1.28e+0)	1.3269e+1 (4.75e-1)	5.1469e+0 (1.78e-1)	5.4711e+0 (2.11e-1)	5.7164e+0 (6.12e-1)	4.9087e+0 (4.98e-1)	4.1744e+0 (1.79e-1)
	10	6.9949e+0 (2.19e-1)	1.1503e+1 (2.78e+0)	1.6456e+1 (6.05e-1)	6.9304e+0 (2.77e-1)	7.3186e+0 (1.38e-1)	7.4463e+0 (8.06e-1)	6.1081e+0 (2.75e-1)	5.9012e+0 (1.83e-1)
	15	1.3467e+1 (4.59e-1)	2.4872e+1 (1.78e+0)	2.8805e+1 (9.17e-1)	1.3371e+1 (3.62e-1)	1.4941e+1 (7.53e-1)	1.7680e+1 (1.36e+0)	1.3177e+1 (1.75e+0)	1.0986e+1 (3.17e-1)
InWFG7	5	2.6606e+0 (2.21e-1)	4.8577e+0 (1.49e+0)	6.7670e+0 (4.00e-1)	2.6622e+0 (1.26e-1)	2.1707e+0 (3.39e-2)	1.7780e+0 (1.10e-1)	1.5382e+0 (1.02e-1)	1.2788e+0 (2.75e-2)
	8	5.2526e+0 (2.24e-1)	9.6463e+0 (1.57e+0)	1.2592e+1 (5.74e-1)	5.2344e+0 (1.82e-1)	5.7140e+0 (3.26e-1)	4.8604e+0 (5.12e-1)	4.6693e+0 (4.04e-1)	4.1411e+0 (1.92e-1)
	10	7.1233e+0 (1.63e-1)	1.0459e+1 (2.46e+0)	1.6435e+1 (7.69e-1)	7.1666e+0 (1.69e-1)	6.9080e+0 (1.34e-1)	7.0029e+0 (1.06e+0)	8.6939e+0 (6.44e-1)	5.8015e+0 (9.84e-2)
	15	1.3558e+1 (5.26e-1)	2.5972e+1 (3.35e+0)	2.7867e+1 (3.21e-1)	1.3547e+1 (3.95e-1)	1.4055e+1 (1.03e+0)	1.6652e+1 (1.49e+0)	1.4368e+1 (7.82e-1)	1.0396e+1 (2.99e-1)
InWFG8	5	2.3661e+0 (1.54e-1)	5.4128e+0 (8.23e-1)	6.5295e+0 (3.16e-1)	2.1328e+0 (2.69e-1)	2.9748e+0 (2.13e-2)	1.6772e+0 (9.68e-2)	1.9903e+0 (1.78e-1)	1.2397e+0 (1.32e-2)
	8	5.3406e+0 (1.89e-1)	1.0803e+1 (1.86e+0)	1.3150e+1 (5.48e-1)	5.0564e+0 (1.99e-1)	5.1485e+0 (3.55e-1)	5.8688e+0 (7.07e-1)	4.3943e+0 (1.46e-1)	3.9741e+0 (1.65e-1)
	10	7.2946e+0 (3.19e-1)	1.2738e+1 (3.07e+0)	1.6685e+1 (3.78e-1)	6.7037e+0 (1.55e-1)	7.2787e+0 (3.24e-1)	7.5028e+0 (6.31e-1)	6.0779e+0 (1.83e-1)	5.6000e+0 (1.28e-1)
	15	1.3818e+1 (6.13e-1)	2.5920e+1 (3.41e+0)	2.8928e+1 (4.63e-1)	1.3029e+1 (4.43e-1)	1.6414e+1 (7.49e-1)	1.7028e+1 (8.06e-1)	1.3094e+1 (1.86e+0)	1.0426e+1 (2.96e-1)
InWFG9	5	1.9345e+0 (1.59e-1)	3.3811e+0 (1.25e+0)	7.1821e+0 (4.33e-1)	1.8485e+0 (1.55e-1)	2.8514e+0 (1.81e-1)	1.3623e+0 (1.26e-1)	1.8497e+0 (1.20e-1)	1.2450e+0 (1.28e-2)
	8	4.8708e+0 (2.67e-1)	1.1253e+1 (2.74e+0)	1.3139e+1 (2.88e-1)	4.9302e+0 (2.99e-1)	4.9801e+0 (1.14e-1)	5.6168e+0 (3.57e-1)	4.4311e+0 (2.29e-1)	4.0423e+0 (1.02e-1)
	10	6.6313e+0 (3.45e-1)	1.2430e+1 (4.43e+0)	1.7511e+1 (6.74e-1)	6.1332e+0 (2.42e-1)	6.8493e+0 (1.25e-1)	7.6026e+0 (5.13e-1)	5.7919e+0 (2.36e-1)	5.2805e+0 (8.73e-2)
	15	1.3485e+1 (4.28e-1)	2.7502e+1 (2.17e+0)	2.7950e+1 (8.38e-1)	1.3045e+1 (4.70e-1)	1.4019e+1 (1.06e+0)	1.6121e+1 (7.34e-1)	1.1583e+1 (5.02e-1)	1.0628e+1 (2.02e-1)
+/-/=		0/34/2	0/35/1	0/36/0	0/34/2	4/30/2	4/30/2	6/24/6	

Table 4

The mean and standard deviation of IIGD obtained by different algorithms on InDTLZ.

Problem	M	IMOEa	SetGA	DI- μ MOGA	InMaOEA	IMOEa/D	EG-IMOEa	IMOEa/D-AWN	IMEA-IED
InDTLZ1	5	1.8141e-1 (8.48e-2)	4.3612e-1 (4.94e-2)	4.0206e-1 (5.49e-2)	2.8779e-1 (1.60e-1)	1.2394e+0 (1.72e+0)	9.1776e-2 (2.03e-2)	1.6064e-1 (1.60e-2)	5.8141e-2 (3.70e-4)
	8	1.1147e+2 (8.03e+1)	5.0398e-1 (1.14e-2)	5.0653e-1 (1.33e-1)	8.7296e+1 (9.28e+1)	2.9534e-1 (4.48e-2)	5.8110e-1 (4.23e-1)	1.8974e-1 (3.41e-3)	1.0919e-1 (1.62e-3)
	10	1.7526e+2 (1.04e+2)	4.7978e-1 (2.60e-2)	6.2441e-1 (2.36e-1)	2.5116e+2 (1.22e+2)	4.1148e-1 (8.55e-2)	4.4777e-1 (1.82e-1)	1.7329e-1 (2.58e-3)	1.1389e-1 (1.71e-3)
	15	1.1799e+2 (7.93e+1)	5.0082e-1 (2.99e-2)	1.4865e+0 (6.90e-1)	2.9063e+2 (8.66e+1)	1.8100e+0 (6.49e-1)	8.5430e-1 (5.20e-1)	1.9263e-1 (1.32e-2)	1.3929e-1 (1.99e-3)
InDTLZ2	5	9.6917e-1 (2.38e-2)	7.3359e-1 (6.96e-2)	8.3847e-1 (4.43e-2)	9.7905e-1 (2.60e-2)	7.0562e-1 (5.48e-3)	6.5744e-1 (3.19e-3)	9.5204e-1 (3.62e-2)	6.5109e-1 (1.64e-3)
	8	1.8873e+0 (1.31e-1)	1.3911e+0 (2.79e-2)	1.6520e+0 (5.45e-2)	1.9390e+0 (1.85e-1)	1.6335e+0 (1.17e-2)	1.4187e+0 (2.15e-2)	1.6793e+0 (4.57e-2)	1.3759e+0 (4.70e-3)
	10	2.0193e+0 (3.71e-2)	1.7346e+0 (3.03e-2)	1.8612e+0 (7.17e-2)	2.0176e+0 (4.46e-2)	2.0479e+0 (7.54e-3)	1.7274e+0 (1.42e-2)	1.9713e+0 (4.98e-2)	1.6849e+0 (4.30e-3)
	15	2.7631e+0 (7.57e-2)	2.4799e+0 (4.19e-2)	3.0190e+0 (6.51e-2)	2.7527e+0 (5.90e-2)	2.8822e+0 (2.85e-2)	2.6941e+0 (5.45e-2)	2.8543e+0 (5.98e-2)	2.4329e+0 (6.87e-3)
InDTLZ3	5	7.3154e-1 (7.24e-1)	1.1560e+0 (4.33e-2)	1.1121e+0 (2.66e-3)	5.0536e-1 (1.78e-1)	2.6281e+1 (1.24e+1)	4.4901e-1 (2.09e-2)	4.9060e-1 (3.42e-2)	1.6497e-1 (1.22e-3)
	8	7.4951e+2 (3.73e+2)	1.2154e+0 (1.19e-3)	8.2146e+0 (3.70e+0)	1.3156e+3 (5.84e+1)	9.6016e-1 (5.53e-1)	1.3787e+1 (6.55e+0)	7.1136e-1 (2.60e-2)	3.4335e-1 (3.49e-3)
	10	1.1338e+3 (2.98e+2)	1.2440e+0 (2.68e-3)	8.2035e+0 (5.93e+0)	1.3787e+3 (1.44e+2)	2.0966e+0 (1.88e+0)	1.8135e+1 (8.51e+0)	7.3462e-1 (3.98e-2)	3.9449e-1 (2.14e-3)
	15	9.7650e+2 (1.99e+2)	1.2979e+0 (3.37e-3)	5.0776e+1 (2.30e+1)	1.4728e+3 (1.34e+2)	1.2907e+1 (7.00e+0)	5.4062e+1 (2.74e+1)	9.8045e-1 (2.10e-1)	5.3570e-1 (3.03e-3)
InDTLZ4	5	3.8966e-1 (3.21e-2)	3.8693e-1 (7.38e-2)	9.3540e-1 (1.14e-1)	3.9194e-1 (8.94e-3)	4.8193e-1 (4.61e-2)	3.9195e-1 (1.05e-1)	5.1703e-1 (3.32e-2)	1.7135e-1 (1.24e-3)
	8	2.3261e+0 (1.24e-1)	6.0784e-1 (5.16e-2)	1.2177e+0 (2.89e-3)	2.3567e+0 (7.04e-2)	6.5149e-1 (4.91e-2)	8.1971e-1 (9.76e-2)	7.5540e-1 (4.82e-2)	3.5419e-1 (3.10e-2)
	10	2.3443e+0 (1.99e-1)	6.2383e-1 (2.88e-2)	1.2647e+0 (1.05e-2)	2.4460e+0 (5.74e-2)	7.2963e-1 (7.57e-2)	6.8957e-1 (6.43e-2)	6.5487e-1 (3.64e-2)	3.9332e-1 (1.15e-3)
	15	2.3871e+0 (3.14e-1)	1.5487e+0 (2.80e-1)	1.7513e+0 (2.17e-1)	2.6984e+0 (3.30e-2)	1.2193e+0 (9.28e-2)	1.0390e+0 (8.26e-2)	8.8838e-1 (5.35e-2)	5.8020e-1 (3.88e-3)
InDTLZ5	5	1.7027e+0 (2.31e-1)	7.3064e-1 (3.64e-2)	5.2733e-1 (1.98e-1)	1.9550e+0 (1.28e-1)	4.5076e-2 (9.77e-3)	2.9906e-1 (7.05e-2)	1.5882e+0 (3.64e-1)	1.0620e-1 (1.57e-2)
	8	2.1516e+0 (3.11e-1)	7.4291e-1 (5.04e-4)	7.6433e-1 (2.50e-1)	2.4726e+0 (3.09e-2)	6.1781e-2 (9.77e-3)	3.1318e-1 (4.30e-2)	1.5263e+0 (3.29e-1)	2.5332e-1 (3.06e-2)
	10	2.1837e+0 (2.20e-1)	7.4252e-1 (3.12e-4)	1.0022e+0 (3.20e-1)	2.4536e+0 (4.08e-2)	8.3721e-2 (1.50e-2)	2.4450e-1 (6.13e-2)	6.1142e-1 (1.08e-1)	2.8907e-1 (9.79e-3)
	15	2.2801e+0 (1.16e-1)	9.1361e-1 (1.35e-1)	1.0274e+0 (1.74e-1)	2.5239e+0 (5.66e-2)	1.5780e-1 (3.86e-2)	3.0122e-1 (6.66e-2)	3.5505e-1 (1.43e-1)	3.5431e-1 (4.53e-2)
InDTLZ6	5	6.2041e+0 (9.05e-1)	7.3152e-1 (5.70e-3)	7.0308e+0 (1.28e+0)	7.1228e+0 (3.20e-1)	3.9100e-2 (4.80e-3)	1.1536e+0 (4.36e-1)	4.0107e+0 (1.08e+0)	8.4876e-2 (1.49e-2)
	8	9.4392e+0 (6.10e-1)	1.2329e+0 (1.03e+0)	7.4481e+0 (1.82e+0)	1.0066e+1 (5.78e-2)	5.8481e-2 (1.33e-2)	1.7359e+0 (1.26e+0)	4.2240e+0 (1.77e+0)	2.0286e-1 (5.71e-2)
	10	9.8801e+0 (2.54e-1)	7.3918e-1 (3.10e-3)	6.3241e+0 (7.21e-1)	1.0072e+1 (4.00e-2)	7.9413e-2 (1.10e-2)	1.4717e+0 (9.97e-1)	3.8227e+0 (9.12e-1)	1.2506e-1 (2.64e-2)
	15	8.7298e+0 (1.15e+0)	3.6957e+0 (6.00e-1)	8.2186e+0 (7.03e-1)	1.0015e+1 (7.85e-2)	1.6107e-2 (7.56e-2)	3.6937e+0 (8.45e-1)	8.8191e-1 (3.96e-1)	7.1206e-1 (3.98e-1)
+/=-/		0/24/0	0/23/1	0/24/0	0/24/0	8/16/0	0/22/2	0/22/2	

Table 5

The mean and standard deviation of Imprecision obtained by different algorithms on InWFG.

Problem	M	IMOEa	SetGA	DI- μ MOGA	InMaOEA	IMOEa/D	EG-IMOEa	IMOEa/D-AWN	IMEA-IED
InWFG1	5	4.4792e-1 (1.38e-1)	1.0671e-1 (2.48e-2)	6.9891e-1 (7.41e-2)	4.5459e-1 (3.88e-2)	5.7347e-1 (1.74e-2)	5.2786e-1 (4.32e-2)	3.6020e-2 (3.20e-2)	1.8404e-1 (2.36e-2)
	8	9.0015e-1 (1.51e-1)	1.0794e-1 (1.56e-2)	1.1032e+0 (9.51e-2)	8.5060e-1 (1.64e-1)	7.2240e-1 (1.26e-1)	6.9586e-1 (1.64e-1)	8.5206e-1 (9.91e-2)	3.8683e-1 (4.91e-2)
	10	1.4120e+0 (1.15e-1)	1.0520e-1 (1.34e-2)	1.4509e+0 (4.43e-2)	1.0244e+0 (6.54e-2)	5.7092e-1 (2.09e-1)	9.6094e-1 (2.37e-1)	5.9052e-1 (2.25e-1)	4.4023e-1 (7.14e-2)
	15	2.4250e+0 (3.64e-1)	1.8097e-1 (5.08e-2)	1.8654e+0 (1.70e-1)	2.0401e+0 (2.88e-1)	8.6501e-1 (3.29e-1)	1.2373e+0 (3.66e-1)	4.0076e-1 (4.31e-1)	9.9076e-1 (2.15e-1)
InWFG2	5	6.0112e-1 (2.95e-2)	1.7858e-1 (3.15e-2)	2.9787e-1 (7.64e-2)	6.0596e-1 (3.24e-2)	7.8807e-1 (1.10e-1)	4.6766e-1 (5.85e-2)	2.7208e-1 (2.79e-2)	8.3100e-2 (6.90e-3)
	8	1.1685e+0 (6.42e-2)	1.5485e-1 (9.46e-2)	1.0812e+0 (3.15e-1)	1.1573e+0 (5.57e-2)	1.0682e+0 (1.72e-1)	9.5049e-1 (1.29e-1)	6.1654e-1 (7.83e-2)	1.0181e-1 (9.93e-3)
	10	1.5647e+0 (4.91e-2)	2.3137e-1 (1.36e-1)	1.8732e+0 (2.65e-1)	1.5772e+0 (7.37e-2)	1.6059e+0 (1.93e-1)	1.1153e+0 (9.49e-2)	6.7826e-1 (8.67e-2)	9.4223e-2 (7.11e-3)
	15	2.7572e+0 (1.80e-1)	7.8001e-1 (5.43e-1)	2.5425e+0 (3.10e-1)	2.8289e+0 (1.74e-1)	2.4266e+0 (3.87e-1)	1.9989e+0 (2.35e-1)	8.3025e-1 (4.73e-1)	1.1406e-1 (2.20e-2)
InWFG3	5	1.1818e+0 (2.79e-2)	8.0909e-1 (3.09e-1)	7.9719e-1 (9.16e-2)	1.1847e+0 (2.27e-2)	1.3086e+0 (1.56e-1)	8.5060e-1 (2.68e-2)	2.0923e-1 (4.05e-2)	2.0923e-1 (2.86e-2)
	8	2.4149e+0 (8.12e-2)	1.8984e+0 (4.83e-1)	1.3780e+0 (4.83e-1)	2.4058e+0 (7.49e-2)	1.9488e+0 (1.37e-1)	1.3278e+0 (1.16e-1)	1.7141e+0 (9.02e-2)	3.2212e-1 (3.64e-2)
	10	3.1682e+0 (6.00e-2)	1.7414e+0 (4.88e-1)	1.3933e+0 (7.62e-1)	3.1474e+0 (4.98e-2)	2.5343e+0 (2.77e-1)	1.7757e+0 (1.20e-1)	2.2753e+0 (1.32e-1)	4.6712e-1 (6.57e-2)
	15	5.1198e+0 (7.91e-2)	2.7447e+0 (1.21e+0)	3.6467e+0 (1.42e+0)	5.0556e+0 (1.45e-1)	4.1263e+0 (2.60e-1)	2.9235e+0 (3.10e-1)	3.0821e+0 (2.15e-1)	1.1362e+0 (3.78e-1)
InWFG4	5	9.0471e-1 (2.79e-2)	2.4941e-1 (2.65e-1)	6.9857e-1 (3.72e-1)	8.4360e-1 (2.63e-2)	9.1520e-1 (3.29e-2)	8.0091e-1 (4.04e-2)	6.3238e-1 (4.11e-2)	1.4165e-1 (1.77e-2)
	8	2.0006e+0 (5.98e-2)	4.3308e-1 (4.89e-1)	2.3768e+0 (3.97e-1)	1.9267e+0 (7.60e-2)	1.2278e+0 (1.47e-1)	1.6411e+0 (1.14e-1)	1.1527e+0 (1.36e-1)	3.0030e-1 (4.64e-2)
	10	2.7941e+0 (1.22e-1)	8.3454e-1 (9.05e-1)	2.8707e+0 (7.88e-1)	2.7801e+0 (9.03e-2)	1.8139e+0 (2.26e-1)	2.0538e+0 (1.66e-1)	1.4861e+0 (8.79e-2)	4.4859e-1 (2.90e-2)
	15	5.3600e+0 (2.63e-1)	1.3511e+0 (1.60e+0)	5.6844e+0 (3.03e-1)	5.2971e+0 (1.35e-1)	5.5404e+0 (1.82e+0)	2.9970e+0 (4.43e-1)	3.7489e+0 (6.90e-1)	7.6565e-1 (9.07e-2)
InWFG5	5	8.9839e-1 (3.16e-2)	3.9529e-1 (2.85e-1)	1.0389e+0 (3.61e-1)	9.5056e-1 (4.15e-2)	8.3285e-1 (8.16e-2)	8.0500e-1 (5.94e-2)	5.0377e-1 (1.15e-1)	1.2340e-1 (1.33e-2)
	8	2.0939e+0 (9.19e-2)	4.7964e-1 (4.75e-1)	2.2544e+0 (7.87e-1)	2.0871e+0 (5.37e-2)	1.7121e+0 (1.23e-1)	1.7383e+0 (1.07e-1)	1.2408e+0 (5.69e-2)	3.9751e-1 (4.39e-2)
	10	2.9261e+0 (1.90e-1)	1.2331e+0 (5.90e-1)	3.4180e+0 (1.33e-1)	3.0118e+0 (1.97e-1)	1.7121e+0 (1.23e-1)	2.2503e+0 (1.53e-1)	1.7211e+0 (9.21e-2)	4.3153e-1 (2.89e-2)
	15	5.8146e+0 (3.42e-1)	1.4827e+0 (1.10e+0)	5.1917e+0 (3.01e-1)	5.7595e+0 (4.14e-1)	3.9444e+0 (5.94e-1)	3.8590e+0 (4.50e-1)	4.1522e+0 (3.66e-1)	9.1577e-1 (6.10e-2)
InWFG6	5	8.6867e-1 (3.67e-2)	4.3325e-1 (2.36e-1)	9.6203e-1 (2.58e-1)	8.4990e-1 (4.35e-2)	8.4098e-1 (8.07e-2)	7.1087e-1 (3.95e-2)	3.8093e-1 (9.73e-2)	1.5810e-1 (1.95e-2)
	8	1.9819e+0 (8.66e-2)	6.1807e-1 (4.41e-1)	1.7533e+0 (6.69e-1)	1.9808e+0 (7.19e-2)	1.5501e+0 (1.29e-1)	1.4699e+0 (9.74e-2)	9.8012e-1 (1.81e-1)	4.1693e-1 (4.62e-2)
	10	2.8470e+0 (1.00e-1)	8.6806e-1 (7.56e-1)	2.9914e+0 (4.42e-1)	2.8734e+0 (9.62e-2)	2.1076e+0 (1.76e-1)	1.8893e+0 (6.99e-2)	1.3935e+0 (1.68e-1)	4.7367e-1 (3.41e-2)
	15	5.4285e+0 (1.75e-1)	7.9017e-1 (9.37e-1)	4.7996e+0 (6.34e-1)	5.4169e+0 (3.63e-1)	5.2260e+0 (4.53e-1)	3.4118e+0 (4.94e-1)	3.3303e+0 (8.42e-1)	8.7912e-1 (9.42e-2)
InWFG7	5	8.3675e-1 (3.02e-2)	3.0914e-1 (2.23e-1)	9.8469e-1 (2.37e-1)	7.9541e-1 (2.90e-2)	8.8419e-1 (7.43e-2)	7.0726e-1 (4.58e-2)	6.6480e-1 (6.16e-2)	1.3392e-1 (1.68e-2)
	8	1.9435e+0 (9.80e-2)	6.8381e-1 (4.44e-1)	1.2163e+0 (5.45e-1)	1.8864e+0 (9.59e-2)	1.3785e+0 (1.75e-1)	1.4122e+0 (1.02e-1)	1.0396e+0 (8.53e-2)	3.9707e-1 (5.86e-2)
	10	2.7934e+0 (7.44e-2)	9.4148e-1 (6.42e-1)	1.2677e+0 (6.93e-1)	2.7775e+0 (7.96e-2)	1.8666e+0 (1.08e-1)	1.9214e+0 (9.29e-2)	5.7481e-1 (4.89e-1)	4.6091e-1 (3.97e-2)
	15	5.1752e+0 (1.59e-1)	1.6472e+0 (1.32e+0)	4.4335e+0 (8.78e-1)	5.0316e+0 (1.34e-1)	6.0471e+0 (1.28e+0)	3.9478e+0 (8.19e-1)	3.0637e+0 (1.72e+0)	1.2701e+0 (2.00e-1)
InWFG8	5	9.7589e-1 (5.27e-2)	2.6436e-1 (9.53e-2)	1.0361e+0 (3.37e-1)	1.0827e+0 (5.89e-2)	9.4803e-1 (3.44e-1)	8.0738e-1 (5.29e-2)	7.8027e-1 (5.46e-2)	1.3991e-1 (9.06e-3)
	8	2.0715e+0 (8.20e-2)	8.6873e-1 (6.57e-1)	1.2319e+0 (5.11e-1)	2.2839e+0 (1.66e-1)	1.6316e+0 (4.72e-1)	1.5789e+0 (1.74e-1)	1.7565e+0 (1.19e-1)	5.4588e-1 (5.49e-2)
	10	2.9455e+0 (1.04e-1)	1.1546e+0 (8.85e-1)	1.3755e+0 (5.36e-1)	3.2827e+0 (1.50e-1)	2.7276e+0 (2.41e-1)	2.0367e+0 (3.20e-1)	2.1824e+0 (2.19e-1)	7.4336e-1 (7.00e-2)
	15	5.5643e+0 (2.76e-1)	1.2838e+0 (1.04e+0)	1.6997e+0 (7.46e-1)	6.1904e+0 (4.30e-1)	6.2724e+0 (4.61e-1)	3.7259e+0 (9.07e-1)	5.1149e+0 (2.52e-1)	7.7349e-1 (1.26e-1)
InWFG9	5	1.0019e+0 (4.16e-2)	2.9556e-1 (2.75e-1)	1.0088e+0 (4.12e-1)	9.8527e-1 (3.73e-2)	9.1909e-1 (1.98e-1)	1.0082e+0 (8.97e-2)	8.9476e-1 (6.16e-2)	2.2240e-1 (2.74e-2)
	8	2.2061e+0 (9.70e-2)	4.9774e-1 (6.18e-1)	2.0346e+0 (9.63e-1)	2.2171e+0 (4.94e-2)	1.8815e+0 (1.93e-1)	2.1485e+0 (1.93e-1)	1.4814e+0 (9.17e-2)	6.9822e-1 (4.11e-2)
	10	3.1896e+0 (3.12e-1)	7.1331e-1 (6.12e-1)	2.9436e+0 (5.71e-1)	3.3497e+0 (1.49e-1)	2.3173e+0 (1.86e-1)	3.0256e+0 (2.52e-1)	1.9260e+0 (1.50e-1)	7.6284e-1 (4.43e-2)
	15	5.9795e+0 (6.73e-1)	1.0237e+0 (1.06e+0)	5.3444e+0 (5.55e-1)	6.4254e+0 (1.97e-1)	5.4978e+0 (2.67e-1)	4.9679e+0 (2.50e-1)	4.7482e+0 (4.98e-1)	1.4507e+0 (1.78e-1)
+/-/=		0/36/0	4/11/21	0/36/0	0/36/0	0/34/2	0/35/1	2/32/2	

Table 6

The mean and standard deviation of Imprecision obtained by different algorithms on InDTLZ.

Problem	M	IMOEa	SetGA	DI- μ MOGA	InMaOEA	IMOEa/D	EG-IMOEa	IMOEa/D-AWN	IMEA-IED
InDTLZ1	5	7.6874e-2 (1.92e-2)	1.6208e-2 (2.05e-2)	8.1929e-2 (2.08e-2)	7.5502e-2 (4.34e-2)	1.5835e+0 (6.01e-2)	6.5370e-2 (7.43e-3)	8.7401e-3 (1.26e-2)	1.9160e-2 (2.30e-2)
	8	5.4580e+1 (3.04e+0)	9.2913e-2 (5.85e-3)	2.3519e-2 (1.88e-2)	2.7564e+1 (2.02e+1)	2.5471e+0 (6.22e-2)	1.4663e-1 (1.25e-1)	5.2703e-2 (5.71e-2)	2.9660e-2 (2.22e-3)
	10	6.6940e+1 (1.65e+0)	9.2575e-2 (6.09e-3)	2.5192e-2 (3.44e-2)	6.6869e+1 (2.13e+0)	3.0637e+0 (5.95e-2)	1.1033e-1 (4.12e-2)	5.7370e-2 (2.03e-2)	2.4572e-2 (2.03e-3)
	15	6.3189e+1 (4.68e+0)	9.3600e-2 (5.70e-3)	1.5485e-1 (1.05e-1)	6.4941e+1 (4.12e+0)	4.5771e+0 (1.32e-1)	2.0546e-1 (1.15e-1)	6.9712e-2 (2.60e-2)	4.8302e-2 (4.35e-2)
InDTLZ2	5	4.6386e-1 (1.07e-2)	1.2893e-1 (1.32e-1)	3.9123e-1 (2.12e-2)	4.6230e-1 (7.74e-3)	1.7132e+0 (2.61e-2)	3.7557e-1 (9.69e-3)	4.0734e-1 (6.28e-2)	1.5306e-1 (6.17e-3)
	8	1.0214e+0 (5.48e-2)	2.4883e-1 (2.42e-1)	7.6457e-1 (2.59e-2)	1.0740e+0 (6.71e-2)	2.6675e+0 (3.63e-2)	7.3044e-1 (1.65e-2)	8.4856e-1 (1.39e-1)	3.6138e-1 (9.83e-3)
	10	1.2942e+0 (2.24e-2)	3.2606e-2 (2.33e-2)	9.7821e-1 (2.20e-2)	1.3015e+0 (3.09e-2)	3.2986e+0 (4.09e-2)	9.5583e-1 (1.82e-2)	1.0156e+0 (4.36e-2)	4.0759e-1 (1.11e-2)
	15	2.0267e+0 (5.57e-2)	6.4029e-1 (3.86e-1)	1.7202e+0 (3.89e-2)	2.0287e+0 (3.65e-2)	4.8149e+0 (7.10e-2)	1.6215e+0 (2.82e-2)	1.5845e+0 (5.61e-2)	3.1979e-1 (2.93e-2)
InDTLZ3	5	3.9016e-1 (2.22e-1)	2.3095e-2 (3.40e-2)	1.8099e-1 (3.44e-2)	1.0621e+0 (8.67e-1)	1.5251e+0 (7.59e-2)	1.3550e-1 (5.79e-2)	1.1895e-2 (1.80e-3)	5.4693e-2 (3.93e-3)
	8	2.7992e+2 (1.09e+1)	1.9159e-1 (6.17e-3)	9.9865e-1 (6.73e-1)	3.4335e+2 (1.42e+1)	2.5526e+0 (4.70e-2)	1.6974e+0 (7.63e-1)	6.2173e-2 (1.02e-1)	2.1098e-1 (2.14e-1)
	10	4.1179e+2 (1.52e+1)	1.8952e-1 (8.32e-3)	1.6532e+0 (1.21e+0)	4.3348e+2 (1.35e+1)	3.0660e+0 (7.37e-2)	2.5897e+0 (1.27e+0)	7.6231e-1 (9.40e-1)	1.8158e-1 (4.72e-2)
	15	4.8400e+2 (4.90e+1)	1.8680e-1 (8.52e-3)	8.0048e+0 (3.25e+0)	5.5620e+2 (2.22e+1)	4.6399e+0 (1.08e-1)	1.2022e+1 (4.85e+0)	3.2409e-1 (1.70e-1)	2.8771e-1 (2.08e-1)
InDTLZ4	5	1.5997e-1 (7.41e-3)	2.1774e-1 (2.82e-3)	1.0767e-1 (4.42e-2)	1.5228e-1 (6.72e-3)	1.6252e+0 (5.77e-2)	1.2867e-1 (2.66e-2)	2.3523e-2 (7.05e-3)	4.7186e-2 (3.71e-3)
	8	6.5696e-1 (3.57e-2)	1.2148e-1 (3.68e-3)	1.8308e-1 (2.27e-2)	6.5171e-1 (1.83e-2)	2.5180e+0 (3.48e-2)	1.8778e-1 (1.23e-2)	6.8849e-2 (5.74e-2)	1.1201e-1 (8.76e-3)
	10	7.5731e-1 (2.24e-2)	2.1133e-1 (3.44e-3)	2.0338e-1 (6.54e-3)	7.5840e-1 (2.29e-2)	3.0334e+0 (8.68e-2)	1.7002e-1 (1.08e-2)	1.2693e-1 (1.14e-2)	1.3434e-1 (8.65e-3)
	15	9.2690e-1 (3.34e-2)	4.2679e-1 (7.01e-2)	2.9629e-1 (6.31e-2)	9.3586e-1 (2.49e-2)	4.4892e+0 (8.02e-2)	1.7856e-1 (2.20e-2)	1.5656e-1 (1.97e-2)	7.7656e-2 (9.21e-3)
InDTLZ5	5	6.1429e-1 (2.28e-2)	1.1460e-1 (5.36e-2)	2.7692e-1 (5.00e-2)	5.4440e-1 (2.25e-2)	1.5783e+0 (5.73e-2)	2.1401e-1 (7.99e-2)	5.9709e-1 (1.48e-1)	2.1939e-1 (2.85e-2)
	8	8.3868e-1 (2.36e-2)	1.0633e-1 (7.04e-2)	2.3925e-1 (7.44e-2)	8.0121e-1 (3.09e-2)	2.5150e+0 (6.43e-2)	1.6236e-1 (2.30e-2)	3.4486e-1 (2.11e-1)	4.7553e-1 (2.98e-2)
	10	9.5571e-1 (4.01e-2)	7.3212e-2 (6.11e-2)	2.3204e-1 (1.20e-1)	9.1435e-1 (1.45e-2)	3.0247e+0 (9.00e-2)	1.5106e-1 (1.77e-2)	3.4724e-1 (3.07e-2)	5.7372e-1 (2.13e-2)
	15	1.1542e+0 (3.23e-2)	7.1076e-2 (1.08e-1)	2.6538e-1 (1.18e-1)	1.0872e+0 (5.34e-2)	4.4953e+0 (1.04e-1)	2.4937e-1 (3.38e-2)	3.4724e-1 (2.28e-2)	4.6863e-1 (5.89e-2)
InDTLZ6	5	1.5382e+0 (1.17e-1)	5.5790e-2 (1.88e-2)	1.1421e+0 (5.89e-1)	1.4453e+0 (7.63e-2)	1.6199e+0 (4.39e-2)	3.0450e-1 (6.51e-2)	1.0322e+0 (6.16e-1)	2.3234e-1 (3.01e-2)
	8	2.4593e+0 (1.61e-1)	4.7224e-2 (1.29e-1)	8.5980e-1 (2.32e-1)	2.5998e+0 (1.12e-1)	2.5220e+0 (7.13e-2)	6.0014e-1 (1.10e-1)	1.2222e+0 (6.82e-1)	4.5155e-1 (1.73e-2)
	10	2.7611e+0 (4.04e-2)	4.8178e-3 (2.84e-3)	9.5935e-1 (5.13e-1)	2.7251e+0 (1.07e-1)	3.0369e+0 (6.81e-2)	6.1167e-1 (1.49e-1)	1.0942e+0 (7.90e-2)	5.2294e-1 (4.84e-2)
	15	3.4529e+0 (2.02e-1)	8.1162e-1 (2.91e-1)	8.1854e-1 (5.47e-1)	3.2842e+0 (1.68e-1)	4.1496e+0 (1.29e+0)	8.3081e-1 (1.26e-1)	5.7580e-1 (8.32e-2)	7.6418e-1 (4.84e-2)
+/-/=		0/24/0	9/9/6	3/18/3	0/24/0	0/24/0	3/18/3	7/12/5	

Table 7

The mean and standard deviation of Spread obtained by different algorithms on InWFG.

Problem	M	IMOEa	SetGA	DI- μ MOGA	InMaOEA	IMOEa/D	EG-IMOEa	IMOEa/D-AWN	IMEA-IED
InWFG1	5	1.3141e+0 (2.54e-1)	9.5337e-1 (7.06e-2)	9.7667e-1 (2.75e-2)	1.1160e+0 (2.57e-1)	1.8780e+0 (9.81e-3)	1.1315e+0 (6.81e-2)	1.3220e+0 (2.39e-1)	6.5979e-1 (5.00e-2)
	8	1.2561e+0 (1.76e-1)	9.6964e-1 (2.24e-2)	9.9254e-1 (2.93e-2)	1.1508e+0 (3.14e-1)	1.2175e+0 (5.90e-2)	1.1115e+0 (7.11e-2)	1.0549e+0 (7.05e-2)	9.2163e-1 (4.64e-2)
	10	1.2796e+0 (1.31e-1)	9.3702e-1 (1.57e-2)	9.3120e-1 (1.73e-2)	1.2755e+0 (8.10e-2)	1.1119e+0 (1.06e-1)	1.1243e+0 (2.38e-2)	9.3550e-1 (1.19e-1)	9.6663e-1 (7.28e-2)
	15	1.1598e+0 (1.02e-1)	9.9807e-1 (7.22e-3)	9.9803e-1 (2.99e-3)	1.0972e+0 (1.04e-1)	1.0948e+0 (6.33e-2)	1.0760e+0 (4.52e-2)	1.1262e+0 (8.65e-2)	1.2266e+0 (9.90e-2)
InWFG2	5	1.5083e+0 (7.92e-2)	1.0024e+0 (3.74e-2)	1.0053e+0 (1.66e-2)	1.4719e+0 (8.53e-2)	1.8775e+0 (2.45e-2)	1.0792e+0 (2.84e-2)	1.2710e+0 (8.36e-2)	5.8382e-1 (3.11e-2)
	8	1.1962e+0 (9.28e-2)	9.9371e-1 (3.27e-2)	1.0002e+0 (7.54e-4)	1.3027e+0 (3.31e-2)	1.1443e+0 (5.33e-2)	1.0722e+0 (3.24e-2)	1.1285e+0 (7.23e-2)	7.9702e-1 (3.36e-2)
	10	1.2368e+0 (7.08e-2)	9.9799e-1 (1.92e-2)	9.9897e-1 (1.71e-3)	1.2470e+0 (5.36e-2)	1.2179e+0 (3.82e-2)	1.0755e+0 (2.89e-2)	1.1366e+0 (7.07e-2)	8.0965e-1 (2.67e-2)
	15	1.1419e+0 (6.87e-2)	9.7057e-1 (4.42e-2)	9.9914e-1 (6.86e-4)	1.1701e+0 (6.13e-2)	1.0577e+0 (2.13e-2)	1.0515e+0 (1.98e-2)	1.0628e+0 (7.59e-2)	9.9677e-1 (5.16e-2)
InWFG3	5	8.7587e-1 (9.22e-2)	1.0153e+0 (1.50e-2)	1.0000e+0 (6.52e-15)	7.9283e-1 (6.76e-2)	1.3732e+0 (3.04e-2)	1.4384e+0 (8.16e-2)	9.3864e-1 (5.49e-2)	4.0280e-1 (2.22e-2)
	8	1.1833e+0 (8.03e-2)	1.0041e+0 (4.48e-3)	1.0001e+0 (2.70e-4)	1.1733e+0 (4.89e-2)	1.3371e+0 (8.00e-2)	1.5383e+0 (8.06e-2)	1.1005e+0 (9.55e-2)	5.0341e-1 (2.88e-2)
	10	1.1727e+0 (3.92e-2)	1.0041e+0 (3.92e-3)	1.0023e+0 (7.07e-3)	1.1647e+0 (2.65e-2)	1.6539e+0 (3.99e-2)	1.4731e+0 (8.13e-2)	9.5703e-1 (5.91e-2)	4.5834e-1 (2.66e-2)
	15	1.2786e+0 (7.07e-2)	1.0002e+0 (5.30e-4)	1.0000e+0 (6.94e-5)	1.2895e+0 (4.30e-2)	1.3935e+0 (8.88e-2)	1.3052e+0 (6.27e-2)	1.3052e+0 (8.54e-2)	7.0562e-1 (3.82e-2)
InWFG4	5	9.4943e-1 (5.08e-2)	1.3432e+0 (1.89e-1)	1.0079e+0 (1.44e-2)	9.4389e-1 (4.95e-2)	1.9021e+0 (2.42e-2)	1.0755e+0 (1.03e-1)	1.2154e+0 (1.21e-1)	2.9991e-1 (1.39e-2)
	8	8.8195e-1 (5.14e-2)	1.1314e+0 (9.78e-2)	1.0045e+0 (2.63e-3)	8.7582e-1 (5.99e-2)	1.3634e+0 (5.04e-2)	1.0790e+0 (5.44e-2)	1.1653e+0 (1.25e-1)	3.8180e-1 (1.57e-2)
	10	8.0672e-1 (5.02e-2)	1.2413e+0 (3.62e-1)	1.0074e+0 (7.09e-3)	7.8391e-1 (4.95e-2)	1.3692e+0 (7.83e-2)	1.0525e+0 (8.57e-2)	1.1253e+0 (8.46e-2)	3.5555e-1 (1.11e-2)
	15	9.0485e-1 (5.04e-2)	1.0244e+0 (2.04e-2)	1.0009e+0 (1.06e-3)	9.1404e-1 (3.94e-2)	1.2357e+0 (7.94e-2)	1.0825e+0 (5.03e-2)	1.0215e+0 (1.07e-1)	4.8642e-1 (1.83e-2)
InWFG5	5	1.0979e+0 (7.96e-2)	1.4259e+0 (1.45e-1)	9.9344e-1 (1.36e-2)	9.4724e-1 (1.76e-2)	1.8709e+0 (2.18e-2)	1.3026e+0 (9.05e-2)	1.3155e+0 (1.51e-1)	3.0514e-1 (9.88e-3)
	8	1.0242e+0 (7.01e-2)	1.2215e+0 (1.79e-1)	1.0000e+0 (2.97e-5)	1.0259e+0 (5.43e-2)	1.4398e+0 (4.05e-2)	1.1855e+0 (6.78e-2)	1.1284e+0 (1.21e-1)	3.6685e-1 (1.85e-2)
	10	9.3978e-1 (6.48e-2)	1.2549e+0 (1.71e-1)	1.0040e+0 (3.38e-3)	9.7571e-1 (8.29e-2)	1.4347e+0 (3.48e-2)	1.1422e+0 (7.32e-2)	9.0473e-1 (8.42e-2)	3.4524e-1 (1.48e-2)
	15	9.8896e-1 (3.94e-2)	1.0435e+0 (3.12e-2)	1.0012e+0 (1.75e-3)	9.9802e-1 (6.02e-2)	1.2306e+0 (7.53e-2)	1.1611e+0 (5.14e-2)	1.1138e+0 (6.66e-2)	5.3174e-1 (2.67e-2)
InWFG6	5	9.7458e-1 (4.63e-2)	1.1616e+0 (1.67e-1)	9.8858e-1 (2.21e-2)	1.0066e+0 (8.25e-2)	1.8938e+0 (1.30e-2)	1.3755e+0 (5.86e-2)	1.3490e+0 (1.89e-1)	2.9907e-1 (1.18e-2)
	8	8.5282e-1 (6.85e-2)	1.0726e+0 (6.99e-2)	1.0041e+0 (4.06e-3)	9.0831e-1 (5.50e-2)	1.4027e+0 (4.50e-2)	1.2088e+0 (5.37e-2)	1.1928e+0 (6.69e-2)	3.6803e-1 (1.37e-2)
	10	7.8533e-1 (8.81e-2)	1.1206e+0 (1.00e-1)	1.0067e+0 (5.63e-3)	7.8678e-1 (7.41e-2)	1.4657e+0 (4.55e-2)	1.1667e+0 (4.03e-2)	1.2182e+0 (1.29e-1)	3.2853e-1 (1.45e-2)
	15	9.4855e-1 (3.74e-2)	1.0387e+0 (5.34e-2)	1.0019e+0 (1.57e-3)	9.4349e-1 (3.31e-2)	1.1422e+0 (5.21e-2)	1.1197e+0 (4.70e-2)	1.1903e+0 (8.81e-2)	4.5793e-1 (2.27e-2)
InWFG7	5	9.6786e-1 (6.44e-2)	1.2148e+0 (1.95e-1)	1.0021e+0 (6.31e-3)	1.0384e+0 (7.72e-2)	1.8539e+0 (4.06e-2)	1.2891e+0 (6.51e-2)	1.2409e+0 (7.72e-2)	3.0315e-1 (1.54e-2)
	8	8.5177e-1 (6.41e-2)	1.0499e+0 (4.24e-2)	1.0040e+0 (7.22e-3)	8.3287e-1 (6.02e-2)	1.2262e+0 (9.87e-2)	1.1829e+0 (8.73e-2)	1.0297e+0 (7.77e-2)	3.5488e-1 (1.67e-2)
	10	7.8337e-1 (4.40e-2)	1.1860e+0 (1.15e-1)	1.0029e+0 (5.27e-3)	7.7402e-1 (5.67e-2)	1.2587e+0 (4.09e-2)	1.1177e+0 (7.39e-2)	1.3660e+0 (8.92e-2)	3.2665e-1 (1.20e-2)
	15	9.4837e-1 (5.16e-2)	1.0158e+0 (2.65e-2)	1.0016e+0 (1.87e-3)	9.5374e-1 (3.97e-2)	1.1682e+0 (3.20e-2)	1.0971e+0 (5.24e-2)	1.2271e+0 (6.68e-2)	4.9650e-1 (2.60e-2)
InWFG8	5	9.6335e-1 (5.95e-2)	1.0980e+0 (1.01e-1)	1.0001e+0 (8.71e-5)	8.6084e-1 (6.33e-2)	1.8929e+0 (2.42e-2)	1.4715e+0 (8.81e-2)	1.2007e+0 (1.21e-1)	3.0637e-1 (1.40e-2)
	8	8.5433e-1 (5.93e-2)	1.0572e+0 (9.44e-2)	1.0040e+0 (8.09e-3)	8.1754e-1 (5.67e-2)	1.4405e+0 (4.14e-2)	1.1832e+0 (7.53e-2)	1.0355e+0 (1.04e-1)	3.7543e-1 (2.38e-2)
	10	8.0507e-1 (4.58e-2)	1.0936e+0 (8.53e-2)	1.0039e+0 (7.55e-3)	7.1034e-1 (4.37e-2)	1.5183e+0 (2.63e-2)	1.1298e+0 (9.21e-2)	9.0974e-1 (5.68e-2)	3.5975e-1 (1.65e-2)
	15	9.5264e-1 (2.47e-2)	1.0163e+0 (2.75e-2)	1.0006e+0 (7.80e-4)	9.0473e-1 (5.85e-2)	1.1184e+0 (4.55e-2)	1.0913e+0 (5.25e-2)	1.1164e+0 (3.22e-2)	4.3920e-1 (1.80e-2)
InWFG9	5	7.5142e-1 (4.15e-2)	1.4265e+0 (4.42e-1)	9.7539e-1 (4.04e-2)	8.3139e-1 (5.30e-2)	1.8077e+0 (8.02e-2)	8.4995e-1 (1.69e-1)	1.3047e+0 (4.80e-2)	2.9920e-1 (1.33e-2)
	8	8.6140e-1 (5.37e-2)	1.0989e+0 (1.54e-1)	9.9898e-1 (1.99e-3)	8.6491e-1 (5.27e-2)	1.3780e+0 (5.15e-2)	1.0172e+0 (1.05e-1)	1.0581e+0 (1.03e-1)	3.7809e-1 (1.88e-2)
	10	8.3546e-1 (4.21e-2)	1.1142e+0 (5.70e-2)	9.9924e-1 (2.61e-3)	7.5089e-1 (3.42e-2)	1.4110e+0 (2.68e-2)	8.8752e-1 (8.80e-2)	8.9491e-1 (5.64e-2)	3.8038e-1 (2.48e-2)
	15	9.5195e-1 (4.83e-2)	1.0178e+0 (2.03e-2)	1.0003e+0 (9.16e-4)	8.4872e-1 (4.19e-2)	1.2186e+0 (7.50e-2)	1.0922e+0 (4.27e-2)	1.1653e+0 (4.78e-2)	6.1871e-1 (3.31e-2)
+/-/=		0/35/1	1/33/2	1/33/2	1/35/0	1/35/0	1/35/0	0/33/3	

Table 8

The mean and standard deviation of Spread obtained by different algorithms on IndTLZ.

Problem	M	IMOEa	SetGA	DI- μ MOGA	InMaOEA	IMOEa/D	EG-IMOEa	IMOEa/D-AWN	IMEA-IED
IndTLZ1	5	1.2230e+0 (3.41e-1) =	1.0236e+0 (2.66e-2) =	1.0959e+0 (2.68e-1) =	1.4083e+0 (3.74e-1) =	1.5253e+0 (6.69e-2) =	9.5447e-1 (9.12e-2) =	1.6259e+0 (1.18e-1) =	6.3737e-1 (6.29e-1)
	8	6.7823e-1 (6.84e-2)	1.0019e+0 (3.65e-3)	1.0012e+0 (3.86e-3)	8.1545e-1 (1.28e-1)	1.6421e+0 (5.37e-2)	1.0748e+0 (1.12e-1)	1.4482e+0 (3.06e-1)	3.9270e-1 (2.35e-2)
	10	8.6898e-1 (5.70e-2)	1.0087e+0 (6.76e-3)	1.0590e+0 (1.77e-1)	9.6909e-1 (4.86e-2)	1.6647e+0 (5.43e-2)	9.4069e-1 (2.68e-2)	1.3940e+0 (3.53e-2)	5.0931e-1 (3.75e-1)
	15	9.5345e-1 (9.75e-2)	1.0046e+0 (6.22e-3)	1.0033e+0 (6.53e-3)	1.0478e+0 (6.58e-2)	1.3829e+0 (9.73e-2)	1.0731e+0 (1.10e-1)	1.2215e+0 (2.43e-1)	8.6038e-1 (4.04e-1)
IndTLZ2	5	8.7759e-1 (3.67e-2)	1.0748e+0 (6.19e-2)	1.0176e+0 (2.47e-2)	8.6434e-1 (3.42e-2)	9.7010e-1 (5.23e-2)	8.7206e-1 (4.77e-2)	1.2916e+0 (5.78e-2)	4.1080e-1 (2.37e-2)
	8	6.1934e-1 (3.51e-2)	9.5921e-1 (1.09e-1)	1.0048e+0 (6.71e-3)	6.5763e-1 (2.57e-2)	6.5402e-1 (1.59e-2)	9.4494e-1 (3.13e-2)	1.2477e+0 (2.70e-2)	6.1482e-1 (3.52e-2)
	10	5.8303e-1 (2.95e-2)	1.0468e+0 (1.90e-2)	1.0189e+0 (1.14e-2)	5.9370e-1 (2.13e-2)	6.3041e-1 (1.18e-2)	7.9606e-1 (2.73e-2)	1.0943e+0 (2.64e-2)	5.6489e-1 (1.81e-2)
	15	7.6098e-1 (4.96e-2)	1.0070e+0 (2.54e-2)	1.0021e+0 (2.55e-3)	7.6686e-1 (5.25e-2)	7.3368e-1 (3.21e-2)	1.0042e+0 (1.98e-2)	9.7392e-1 (3.14e-2)	7.1007e-1 (1.04e-2)
IndTLZ3	5	1.4968e+0 (3.28e-1)	1.0188e+0 (3.52e-2)	1.0000e+0 (6.93e-5)	1.8225e+0 (6.89e-2)	1.4499e+0 (8.16e-2)	1.7117e+0 (2.03e-1)	1.5468e+0 (1.04e-1)	1.1962e-1 (5.51e-2)
	8	7.5853e-1 (7.70e-2)	1.0004e+0 (7.33e-4)	1.0179e+0 (4.53e-2)	7.5228e-1 (2.55e-2)	1.4800e+0 (6.87e-2)	1.0004e+0 (1.26e-3)	1.4380e+0 (2.17e-1)	3.5345e-1 (4.81e-1)
	10	8.4519e-1 (7.22e-2)	1.0004e+0 (3.25e-4)	1.0006e+0 (1.15e-3)	7.9952e-1 (1.25e-1)	1.6279e+0 (4.89e-2)	1.0176e+0 (2.38e-2)	1.8737e+0 (1.81e-1)	3.3002e-1 (4.65e-1)
	15	9.7243e-1 (1.01e-1)	1.0004e+0 (4.47e-4)	1.0010e+0 (2.59e-3)	9.2946e-1 (1.04e-1)	1.3232e+0 (6.91e-2)	1.0360e+0 (5.65e-2)	1.2680e+0 (3.02e-1)	5.2451e-1 (6.65e-1)
IndTLZ4	5	1.2522e+0 (4.53e-2)	1.4269e+0 (1.13e-1)	1.0199e+0 (2.61e-2)	1.2596e+0 (8.70e-2)	1.5465e+0 (7.00e-2)	1.4932e+0 (1.69e-1)	1.5128e+0 (1.53e-1)	9.1522e-2 (1.28e-2)
	8	9.2587e-1 (8.17e-2)	1.3521e+0 (6.39e-2)	1.0003e+0 (5.58e-4)	1.0044e+0 (6.44e-2)	1.2713e+0 (4.64e-2)	1.0747e+0 (7.08e-2)	1.3395e+0 (8.94e-2)	1.6311e-1 (3.63e-2)
	10	6.8744e-1 (7.66e-2)	1.4470e+0 (6.92e-2)	1.0000e+0 (1.16e-12)	8.3806e-1 (1.18e-1)	1.3698e+0 (6.22e-2)	1.1920e+0 (8.63e-2)	1.4714e+0 (6.98e-2)	1.2749e-1 (1.29e-2)
	15	9.5931e-1 (3.72e-2)	9.5955e-1 (6.26e-2)	1.0000e+0 (0.00e+0)	6.8388e-1 (7.79e-2)	1.0930e+0 (4.22e-2)	1.0995e+0 (4.79e-2)	1.2206e+0 (5.96e-2)	9.2322e-2 (9.31e-3)
IndTLZ5	5	7.5819e-1 (4.31e-2)	1.0033e+0 (8.40e-3)	1.0340e+0 (2.54e-2)	9.3701e-1 (3.87e-2)	1.5994e+0 (4.68e-2)	1.8072e+0 (8.14e-2)	1.3238e+0 (1.34e-1)	6.7364e-1 (4.52e-2)
	8	8.2960e-1 (5.50e-2)	1.0005e+0 (5.53e-4)	1.0000e+0 (3.91e-9)	8.7712e-1 (3.67e-2)	1.5916e+0 (5.94e-2)	1.6602e+0 (9.15e-2)	1.3062e+0 (1.09e-1)	3.8893e-1 (2.33e-2)
	10	7.8269e-1 (4.15e-2)	1.0012e+0 (1.36e-3)	1.0021e+0 (6.52e-3)	8.3253e-1 (3.19e-2)	1.6884e+0 (5.99e-2)	1.6475e+0 (5.33e-2)	1.3946e+0 (3.77e-2)	3.5137e-1 (3.41e-2)
	15	8.9386e-1 (1.75e-2)	1.0021e+0 (1.07e-3)	1.0005e+0 (1.41e-3)	9.4735e-1 (3.01e-2)	1.4876e+0 (1.93e-1)	1.4695e+0 (8.98e-2)	1.3835e+0 (8.12e-2)	5.8645e-1 (8.88e-2)
IndTLZ6	5	9.0000e-1 (5.04e-2)	1.0020e+0 (1.22e-3)	1.0020e+0 (4.18e-3)	1.0051e+0 (3.54e-2)	1.5679e+0 (4.91e-2)	1.4401e+0 (1.26e-1)	1.3411e+0 (9.46e-2)	8.2724e-1 (6.78e-2)
	8	9.9524e-1 (3.19e-2)	1.0041e+0 (5.91e-3)	1.0009e+0 (2.95e-3)	1.0455e+0 (4.09e-2)	1.4727e+0 (5.84e-2)	1.2958e+0 (1.54e-1)	1.4423e+0 (1.24e-1)	5.1933e-1 (1.04e-1)
	10	9.9539e-1 (2.29e-2)	1.0005e+0 (5.46e-4)	1.0038e+0 (1.21e-2)	1.0044e+0 (3.04e-2)	1.5709e+0 (5.68e-2)	1.3188e+0 (9.67e-2)	1.3975e+0 (3.67e-2)	4.8489e-1 (6.05e-2)
	15	9.6577e-1 (4.74e-2)	1.0013e+0 (8.85e-4)	1.0003e+0 (6.28e-4)	1.0196e+0 (3.63e-2)	1.4931e+0 (2.26e-1)	1.1344e+0 (6.02e-2)	1.4474e+0 (1.32e-1)	4.9277e-1 (5.25e-2)
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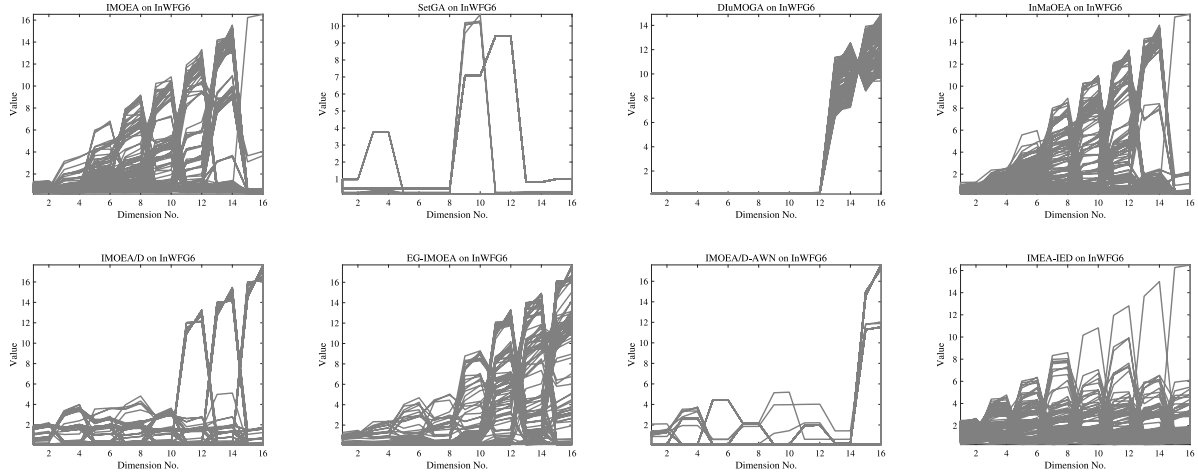


Fig. 5. Results obtained by the 8 algorithms on InWFG6 problems with 8-objective.

Table 9

The average Runtime and runtime rank of different algorithms on two types of benchmarks.

Problem	InWFG	InDTLZ	Rank
IMOEA	5.10E+03	1.00E+04	6
SetGA	6.03E+02	6.20E+02	2
DI-μMOGA	5.08E+03	9.75E+03	5
InMaOEA	4.71E+03	1.04E+04	7
IMOEA/D	4.60E+02	4.30E+02	1
EG-IMOEA	6.80E+03	1.46E+04	8
IMOEA/D-AWN	4.82E+03	6.30E+03	4
IMEA-IED	9.97E+02	2.46E+03	3

Table 10

The average rankings of IDM (Convergence), IDM (Uncertainty), IDM (Without), and IDM (With IED).

Algorithm	Average ranking	
	IIGD	Imprecision
IDM (Convergence)	2.25	3.39
IDM (Uncertainty)	2.78	1.55
IDM (Without)	2.72	3.33
IDM (With IED)	2.25	1.72

Table 11

The average rankings of IDM 1, IDM 2, IDM 3, IDM 4, and IDM (With IED).

Algorithm	Average ranking	
	IIGD	Imprecision
IDM 1	3.14	2.16
IDM 2	2.58	2.47
IDM 3	2.86	4.08
IDM 4	4	4.63
IDM (With IED)	2.41	1.63

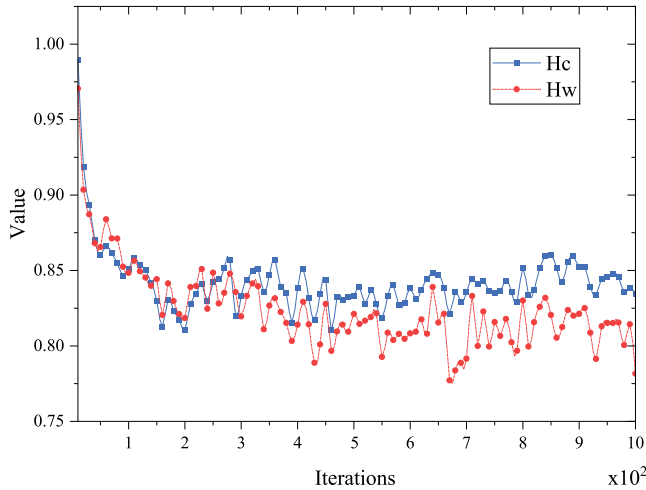


Fig. 6. Dynamic display of convergence entropy and uncertainty entropy for the last objective in the 8-objective InWFG4.

space. The diversity of DI-μMOGA is inferior to other algorithms. This may be because elite archiving-based methods in IMAOPs hinder the entry of new solutions into the population. Combining Tables and Fig. 5, it is evident that IMEA-IED demonstrates excellent convergence and diversity.

4.4.3. Effectiveness of interval dominance method based on information entropy

This section will discuss the role of information entropy in interval comparison, as well as the efficacy of the dominance methodology proposed in Section 3. Fig. 6 illustrates the dynamic changes of convergence entropy (H_c) and uncertainty entropy (H_w) for the last objective

in the 8-objective InWFG4. It can be observed that the values of H_c and H_w fluctuate throughout the iteration process, demonstrating the adaptability of the information entropy dominance method. In generations 1–220, H_c is often smaller than H_w , leading to the utilization of convergence-based interval confidence for interval comparison. After generation 300, H_w becomes smaller than H_c , which results in the emphasis on uncertainty-based interval confidence for interval comparison. This suggests that in the early stages of evolution, due to the unstable convergence of intervals, the dominance method enhances convergence selection. In contrast, in the later stages, our method focuses on improving uncertainty.

An ablation experiment is conducted to demonstrate the effectiveness of the information entropy dominance method. This experiment compares four methods. The first method is the information entropy dominance method proposed in this paper, denoted as IDM (With IED). The method lacks adaptive adjustment of information entropy and only uses convergence-based interval confidence, denoted as IDM (Convergence). The third is without adaptive adjustment of information entropy and only uses uncertainty-based interval confidence, denoted as IDM (Uncertainty). The fourth is the traditional IDM 3 without any improvements, denoted as IDM (Without). These dominance methods are compared in terms of IIGD and Imprecision within the framework of InMaOEA.

A statistical analysis of the average rankings of the algorithms on InWFG is conducted, and the results are presented in Table 10. IDM

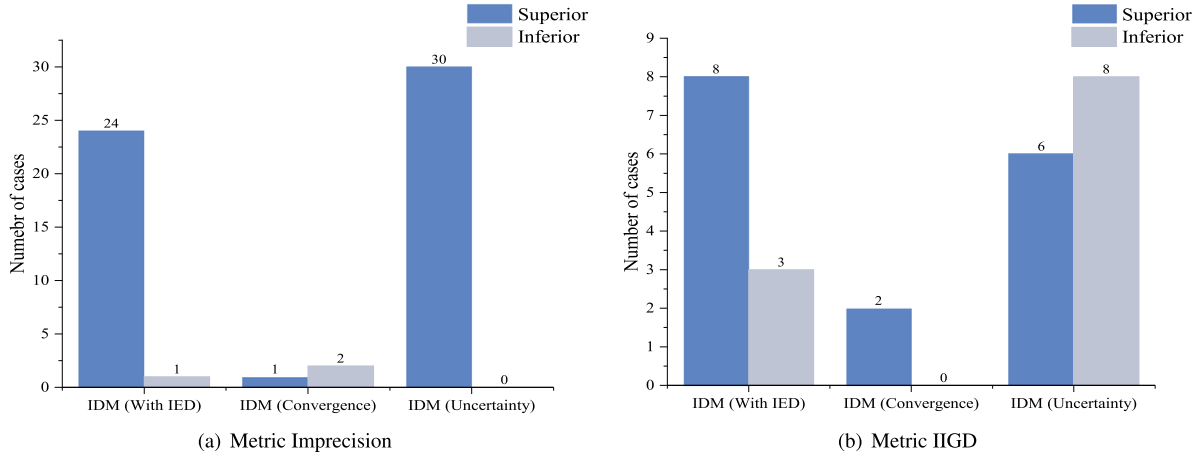


Fig. 7. Wilcoxon rank-sum test analysis between IDM (Convergence), IDM (Uncertainty), IDM (With IED) and IDM (Without).

Table 12

The average rankings of IMEA-IED, variant 1 (With CDI), variant 2 (With AUV), and variant 3 (Without).

Algorithm	Average ranking	
	IIGD	Spread
Variant 1 (With CDI)	2.38	1.91
Variant 2 (With AUV)	2.67	3.06
Variant 3 (Without)	2.8	3.29
IMEA-IED	2.15	1.73

(Convergence) ranks best in IIGD but worst in the Imprecision. IDM (Uncertainty) ranks best in the Imprecision but worst in the IIGD metric. IDM (With IED) outperforms the traditional method IDM (Without) in both IIGD and Imprecision, demonstrating the effectiveness of the proposed information entropy dominance method. Additionally, the Wilcoxon rank-sum test is used to examine the differences between the three dominance methods and IDM (Without). As shown in Fig. 7(a), IDM (Uncertainty) significantly outperforms IDM (Without) in 30 cases in terms of the Imprecision, proving that uncertainty-based interval confidence can enhance uncertainty. However, in the IIGD metric, the number of cases where IDM (Uncertainty) is inferior to IDM (Without) is more than the number of superior cases. Similarly, IDM (Convergence) cannot outperform IDM (Without) in both metrics. In contrast, the proposed IDM (With IED) shows advantages over IDM (Without) in both metrics, indicating that the proposed information entropy dominance method can comprehensively improve convergence and uncertainty.

Table 11 compares the mean rankings of IDM (With IED) with the four dominance methods presented in Section 2.3. It can be seen that IDM (With IED) outperforms the other four methods of both metrics. This demonstrates the superiority of the information entropy-based interval dominance method.

4.4.4. Effectiveness of adaptive niche selection strategy

The adaptive niche selection strategy includes the crowding distance increment (CDI) method and the adaptive update of reference vectors (AUV) method. An ablation experiment is conducted to validate the effectiveness of the CDI and AUV methods. This experiment compares IMEA-IED with three variants. Variant 1 (With CDI) retains the proposed CDI method but does not implement the AUV method. Variant 2 (With AUV) retains the proposed AUV method but replaces the CDI method with the traditional niche count selection. Variant 3 (Without) does not use the AUV method or the CDI method, and the niche selection is the same as NSGA-III.

A statistical analysis of the average rankings of these algorithms in terms of IIGD and Spread metrics is conducted, and the results are

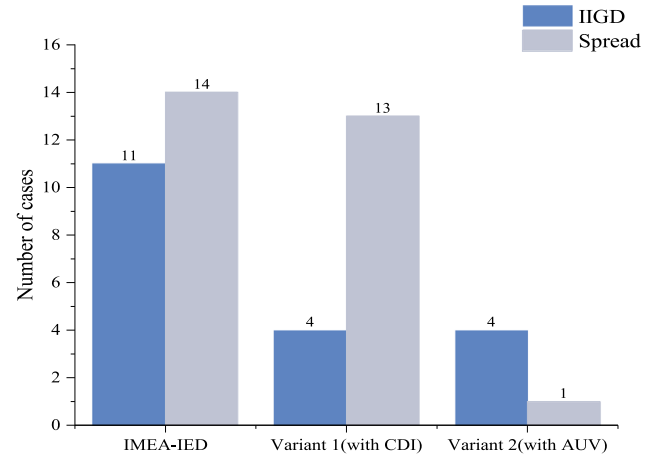


Fig. 8. The number of IMEA-IED, variant 1 (With CDI), and variant 2 (With AUV) with better metrics than variant 3 (Without).

presented in Table 12. The rankings of Variant 1 (With CDI) and Variant 2 (With AUV) are superior to Variant 3 (Without). IMEA-IED achieves the best rankings in both IIGD and Spread. The Wilcoxon rank-sum test is performed to compare the three improved methods with Variant 3 (Without). The blue bars in Fig. 8 represent the number of cases where algorithms outperform Variant 3 (Without) in terms of the IIGD metric, while the red bars represent the number of superior cases in terms of the Spread metric. Both Variant 1 (With CDI) and Variant 2 (With AUV) can improve IIGD and Spread, indicating that the CDI method and the AUV method can enhance the diversity of the algorithm. IMEA-IED significantly outperforms Variant 3 (Without) in the highest number of cases, demonstrating that the combination of the CDI method and the AUV method can further improve the performance of the algorithm.

4.4.5. Application to uncertain workflow migration optimization problem

In edge computing, the edge servers process tasks sent by mobile devices. However, mobile devices may exceed the coverage range of edge servers and require task migration to ensure service continuity. For the workflow migration problem of multiple mobile devices and edge servers, Shi et al. transformed uncertain factors such as network bandwidth and server computing power into interval parameters and proposed an interval many-objective optimization workflow migration model in uncertain environments (IMAOWMUE) [28]. The four objectives of the model are: (1) migration delay; (2) maximum completion time; (3) energy consumption; (4) load balancing.

Table 13
Comparison of different algorithms in practical applications.

Algorithm	HV	Imprecision	Spread	Runtime
IMOEa	4.36e+00	2.64e+00	1.02e+00	3.25e+03
SetGA	3.49e+00	2.29e+00	1.01e+00	3.60e+03
DI- μ MOGA	3.26e+00	3.11e+00	9.63e-01	1.21e+04
InMaOEa	3.37e+00	2.99e+00	9.99e-01	1.08e+04
IMOEa/D	2.80e+00	2.85e+00	1.00e+00	2.92e+04
EG-IMOEa	3.84e+00	2.70e+00	9.94e-01	9.28e+03
IMOEa/D-AWN	3.28e+00	2.32e+00	1.01e+00	3.69e+04
IMEA-IED	4.53e+00	2.30e+00	9.49e-01	2.98e+03

In this section, the algorithm is applied to the workflow migration problem in edge computing and compared with other algorithms to prove the effectiveness of the proposed algorithm. The population size is set to 120 and the number of iterations is set to 100. The other experimental settings for all algorithms are the same as those in Section 4.3. Due to the unknown Pareto front of the actual problem, we used a hypervolume (HV) metric [55] to evaluate the convergence and diversity of algorithms. The larger the HV, the better the performance. The reference point for calculating the hypervolume is [2, 2, 2, 2].

As shown in Table 13, this algorithm ranks first in the three indicators of HV, Spread, and Runtime. SetGA ranks first in the Imprecision. Among the remaining algorithms, IMEA-IED has the best Imprecision. This further demonstrates that the method proposed in this paper can effectively balance convergence, uncertainty, and diversity. In addition, IMOEa/D and SetGA have a longer Runtime, which is different from their performance in the benchmark problems with a smaller Runtime. This may be due to the long calculation time of IMaOWMUE's objective function, resulting in the runtime of decomposition-based and set-based algorithms being longer than that of dominance-based algorithms. After the above comparison, it can be concluded that IMEA-IED can effectively solve the workflow migration problem.

5. Conclusion

Solving the IMaOPs is challenging, but there is limited research attention on this issue. This paper proposes an adaptive many-objective optimization algorithm with information entropy dominance to address this problem. The interval dominance method based on information entropy is proposed to adaptively improve the convergence and uncertainty of the algorithm. In addition, compared to the original niche selection strategy, the crowding distance increment is designed to enhance subpopulation selection. An adaptive strategy for updating reference vectors is also proposed based on association information. The experimental results on interval benchmark problems and a workflow migration problem demonstrate that IMEA-IED can effectively solve IMaOPs. However, in this paper, interval information entropy is compared in each generation, and the reference vectors are updated in each generation. This increases some runtime. Future work will analyze the evolutionary process of the population, enhance the corresponding selection ability at appropriate stages, and reduce the algorithm runtime.

CRedit authorship contribution statement

Zhihua Cui: Project administration, Methodology. **Conghong Qu:** Writing – original draft, Methodology, Investigation. **Zhixia Zhang:** Methodology. **Yaqing Jin:** Methodology. **Jianghui Cai:** Supervision. **Wensheng Zhang:** Supervision. **Jinjun Chen:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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